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January 29 - February 2, 2018 · Barcelona

BRKARC-3873

Catalyst 9400 Architecture

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Cisco Spark

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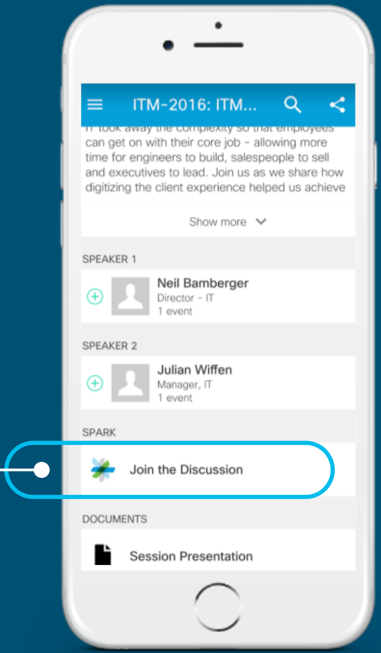

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Session Goal

- To provide a thorough understanding of the Catalyst 9400 hardware architecture, packet flows and key forwarding components
- This session will also examine Catalyst 9400 software architecture, ACL and QoS
- No discussion on other Catalyst platforms

Agenda

- Overview
- Architecture
- Forwarding
- ACL
- QoS
- High Availability
- Conclusion

Overview

A New Era in Networking

Previous Era

New Era

Software Defined Access
(SD-Access)

(Catalyst 9K Series)

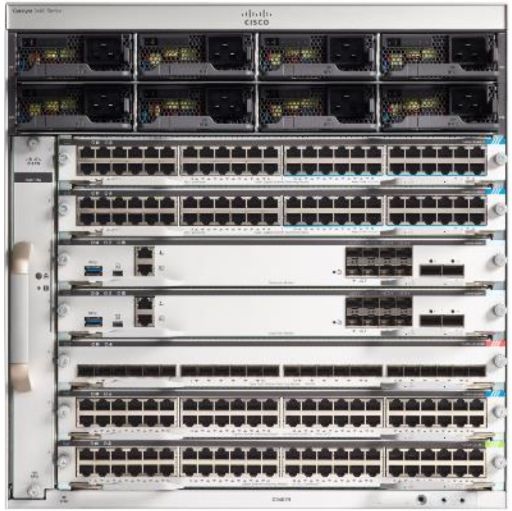
SD-Access - *Policy Based Automation from Edge to Cloud*

Catalyst 9400

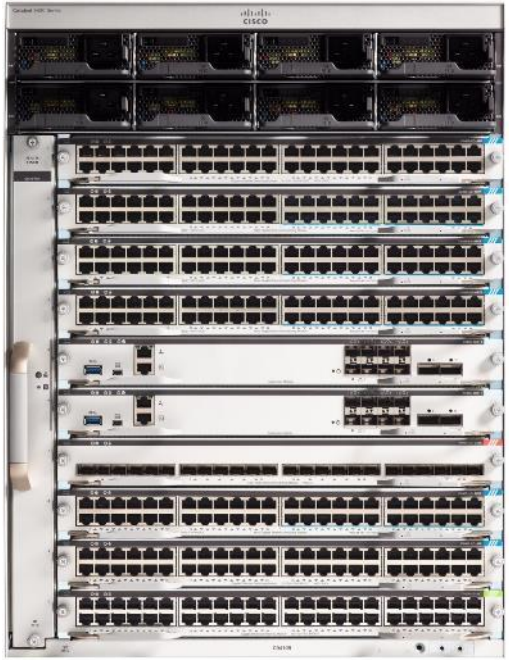
480G BW per slot

Redundancy is now Table-stakes

Ready for 100W PoE



7-Slot



10-Slot

Modular Access Platform

Hardware

Chassis

	7 - Slot	10 - Slot
Supervisor	2 (Redundant)	
Line Cards	5	8
Ports	240x 10/100/1000 120 mGig; 128 SFP/SFP+ 2x QSFP+	384x 10/100/1000 192 mGig; 200 SFP/SFP+ 2x QSFP+
Dimension	W:17.5”; D:16.25”; H:10RU	W: 17.5”; D:16.25”; H: 13RU
BW per LC Slot	480G	480G
BW between Sup Slots	720G	
Power Supply	8 PS (N+1 and N+N)	
PoE per slot	4,800W	
Cooling	Side to Side (Front-to-Back for PS)	

High Density 10G Ports, 100G Uplinks

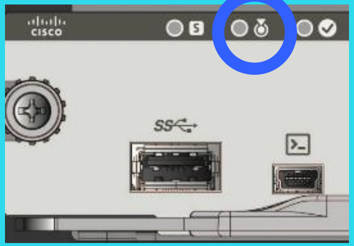
Ready for future higher power PoE devices

Blue Beacon

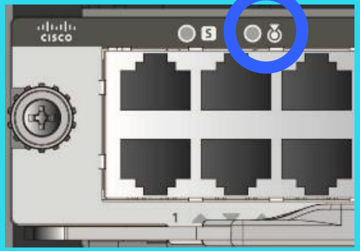
- Chassis/Fan tray, Supervisor, Line cards, and PS have the beacon
- Identify the hardware during configuration and trouble shooting
- Turn on and off by using software control (No button except FanTray)

Chassis/FanTray

Supervisor



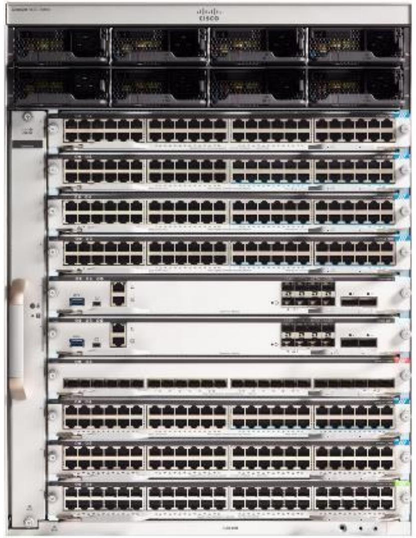
Line Card



PS

RFID

RFID on Every FRUable Components of Catalyst 9400



Built-in Passive RFID

RFID

Sample RFID Tag Data

SN	= 'FOC2109Q023’
PID	= 'C9410R'
VID	= 'V00'
TAN	= '68-100900-02'
TAN Rev	= '10'
CLEI	= 'UNDEFINED'
Index	= '900'
Encode	= 'SGTIN-198‘
Filter	= '0‘
Partition	= '5'
Company	= '0746320'

Inventory Management (Tracking) has never been Easier

Sup-1 - Overview

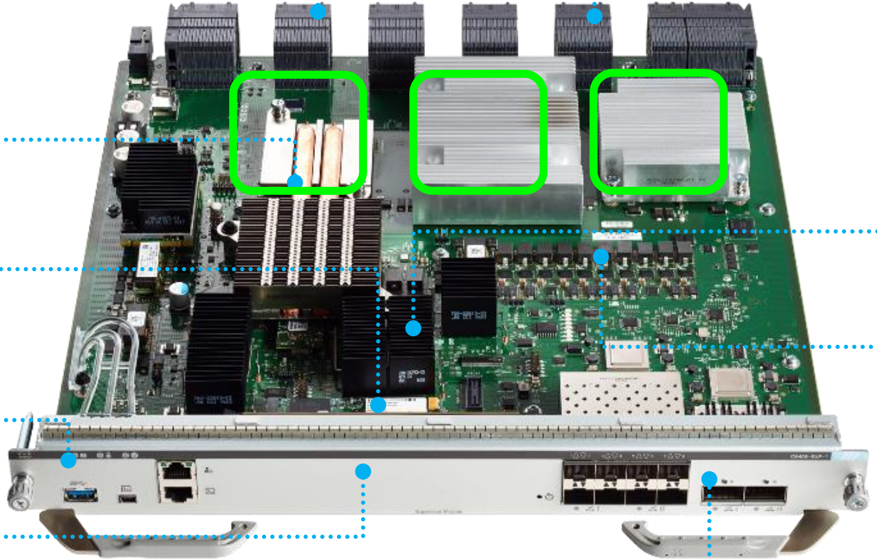
720G LCs/Uplinks

UADP 2.0 XL ASICs

M.2 SATA SSD
(Optional: Upto 1TB)

USB 2.0/3.0

MACSec256



Line Card Slot BW:
7 Slot: 80G
10 Slot: 80G (> 150Byte)

2.4Ghz Quad Core x86 CPU

16G DRAM and 16G Flash

Uplinks:
8x 10G / 2x 40G

Sup-1XL - Overview

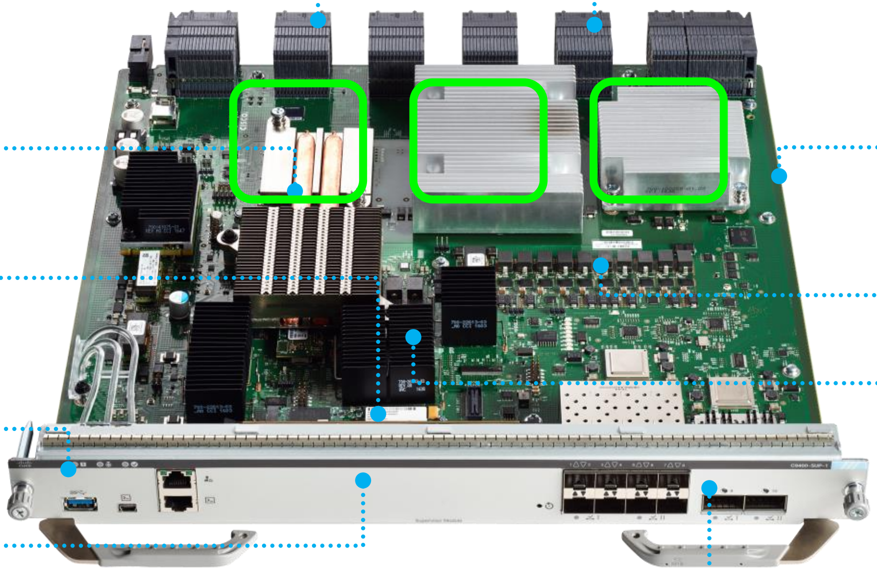
720G LCs/Uplinks

UADP 2.0 XL ASICs

M.2 SATA SSD
(Optional: Upto 1TB)

USB 2.0/3.0

MACSec256



Line Card Slot BW:
7 Slot: **120G (>150Byte)**
10 Slot: 80G (> 150Byte)

Templates: Distribution, Core, SD Boarder, NAT

16G DRAM and 16G Flash

2.4Ghz Quad Core x86 CPU

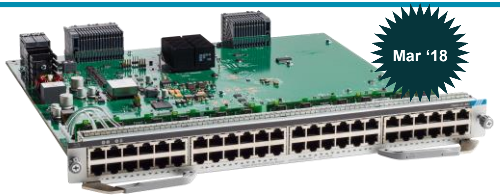
Uplinks:
8x 10G / 2x 40G

Optimized for Core deployment

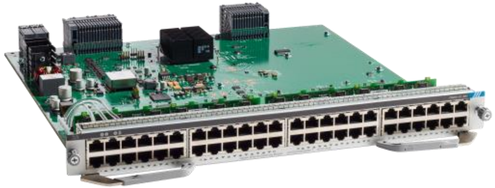
Line Cards - Copper

48x 10/100/1000 Data

RJ45 (Data)
48x 10/100/1000
TrustSec and MACSec(256)

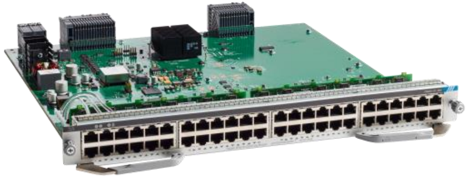


48x 10/100/1000 PoE/PoE+



48x 10/100/1000 PoE/Poe+/UPoE

RJ45 (UPoE)
48x 10/100/1000
PoE/PoE+; PoE/PoE+/UPoE
TrustSec and MACSec(256)



24x 1G + 24x mGig UPoE

RJ45 (mGig)
24x 10/100/1000 + 24x 100/1G/2.5G/5G/10G
PoE/PoE+/UPoE
TrustSec and MACSec(256)

Line Cards - Fiber

24x SFP

48x SFP

SFP (1G)
48x 100/1000
TrustSec and MACSec(256)

24x SFP/SFP+

Fiber (1G/10G)
24x 1G/10G
TrustSec and MACsec(256)

Power Supplies

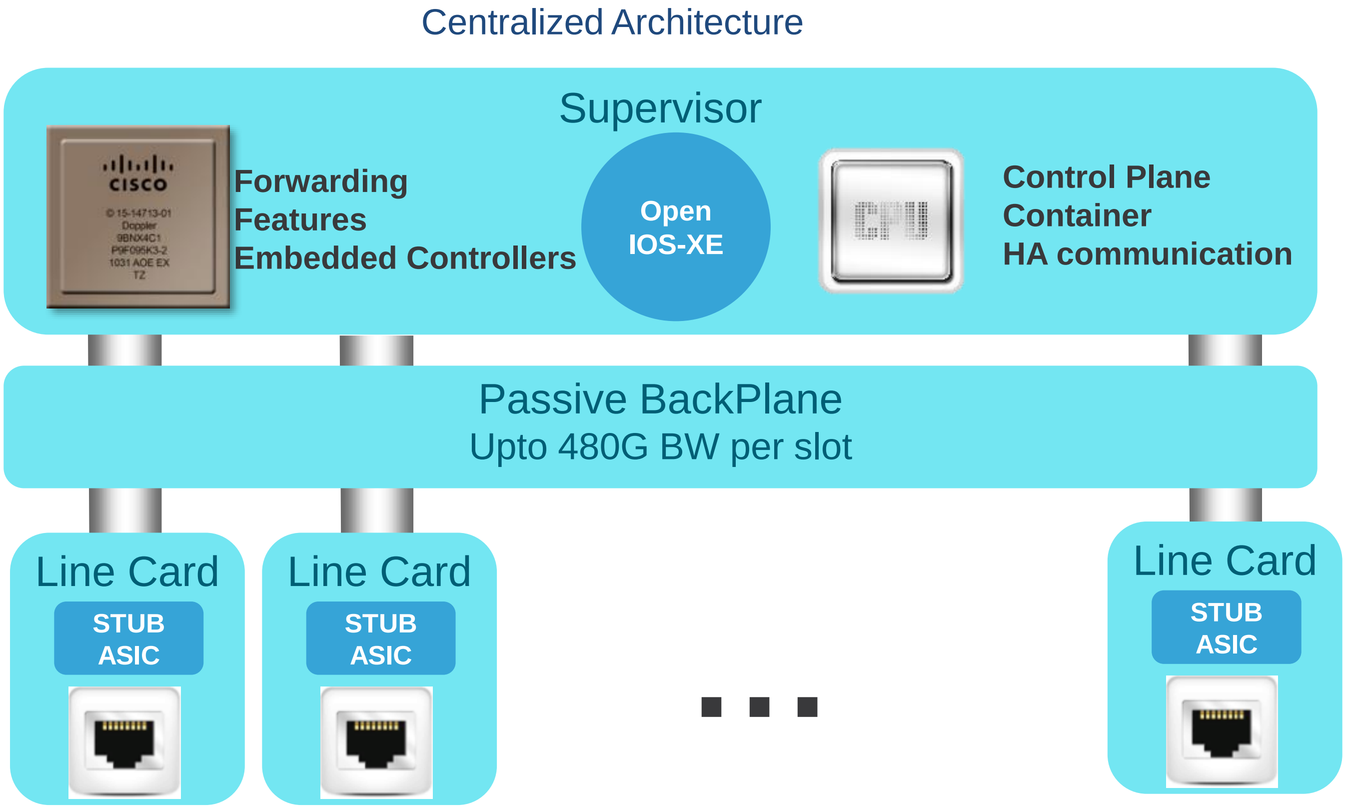
- Modular Design: 8 PS for 7 and 10 slot chassis
- Shared: Power for both Data and Inline Power
- Redundancy Mode : N+N , N+1
- Platinum PS: 90%+ efficiency
- Output: 3200W AC PS With 240V input. (1570W with 120V input)

Fan Tray

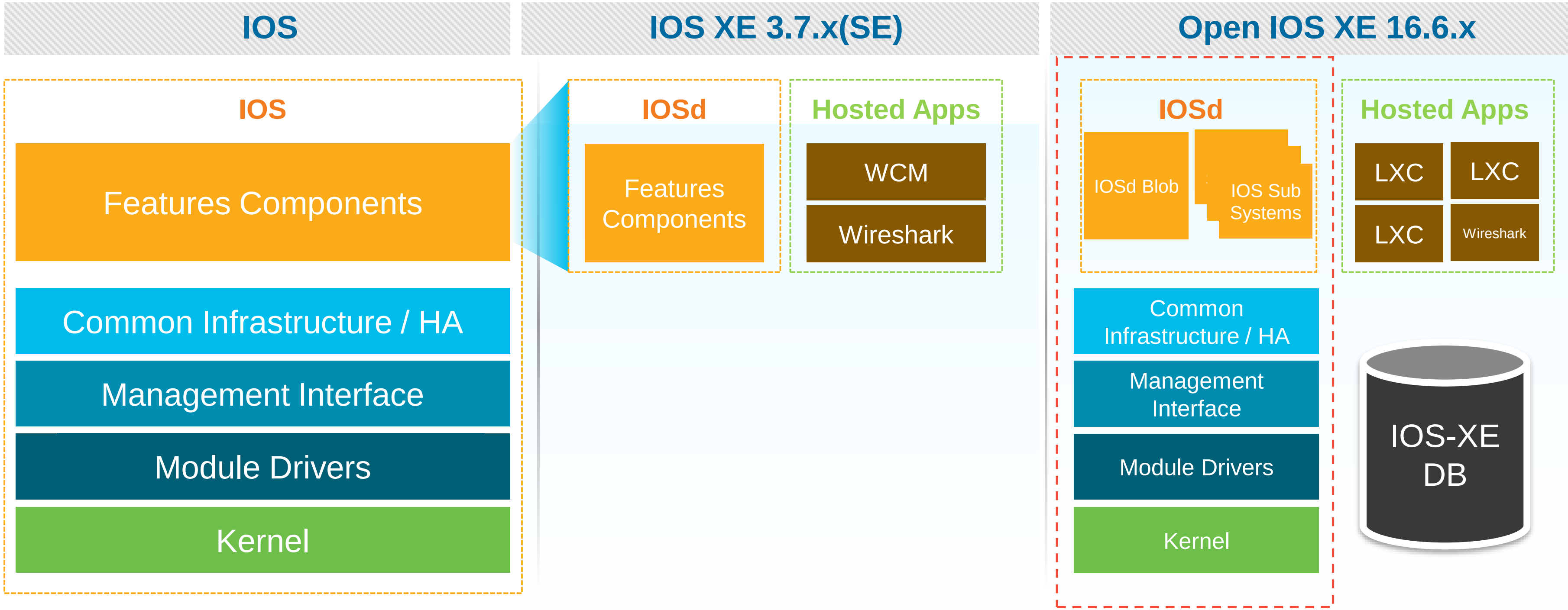
- Redundant - N+1 fan
- Flexible Service - fan tray can be replaced from the portside or the back
- Efficient - Variable speed per fan depends on the load, temperature and altitudes (=>lower noise).
- Air flow - Side to side air flow

Architecture

Architecture

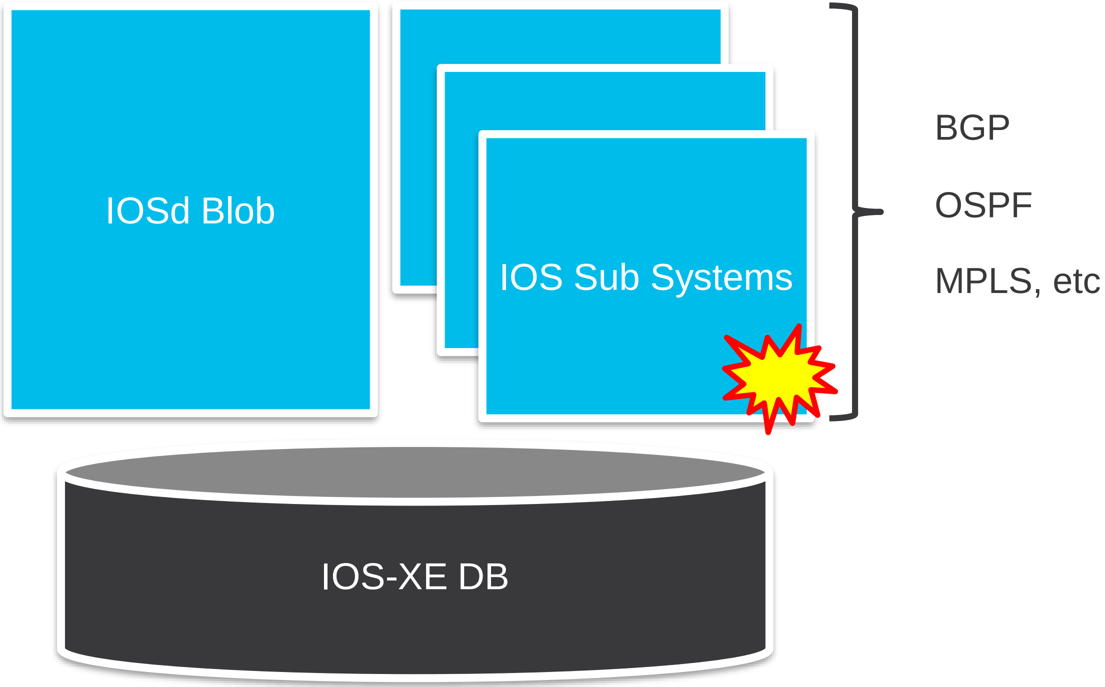


Open IOS-XE



Same Look & Feel, Enhanced & Modern Architecture

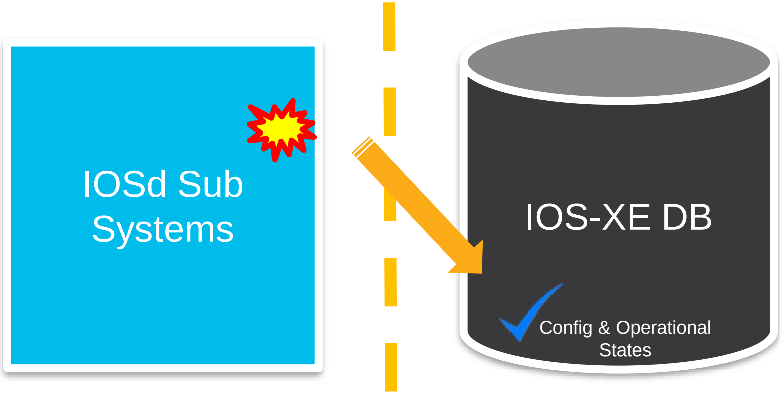
Open IOS XE – IOS Sub Systems



Failure of one of the Sub Systems Keeps Rest of the System intact

IOS Sub Systems Enhances IOS Resiliency

Open IOS XE – DB



Decoupling Code & Data protects the Operational & Configurational States

Higher Application UP Time

Quicker Recovery

Better Convergence

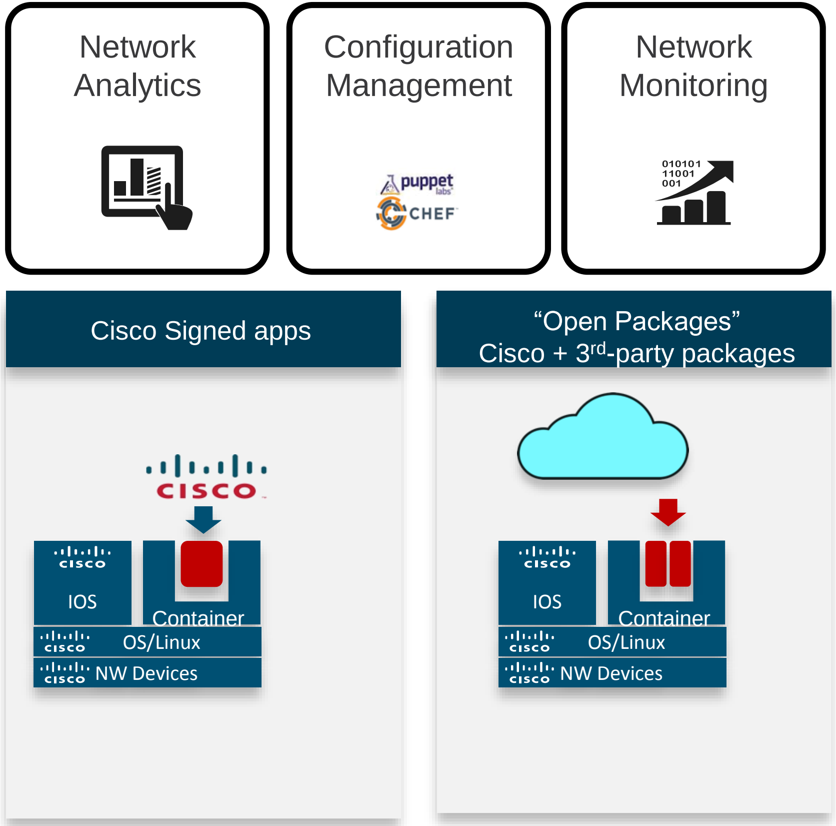
IOS-XE – An Application Platform

Kernel Support for Multiple Containers exist in current versions of IOS-XE Denali

Depending on the Platform Capabilities, Apps can run in Containers

netconf/restconf/yang/rest-api Interfaces

Life Cycle Management



UADP 2.0 - Next Generation of ASIC Innovation

Investment Protection
Flexible Pipeline

Universal Deployments
Adaptable Tables

Enhanced Scale/Buffering
Multicore resource share

384K Flex Counters

Shared Lookup

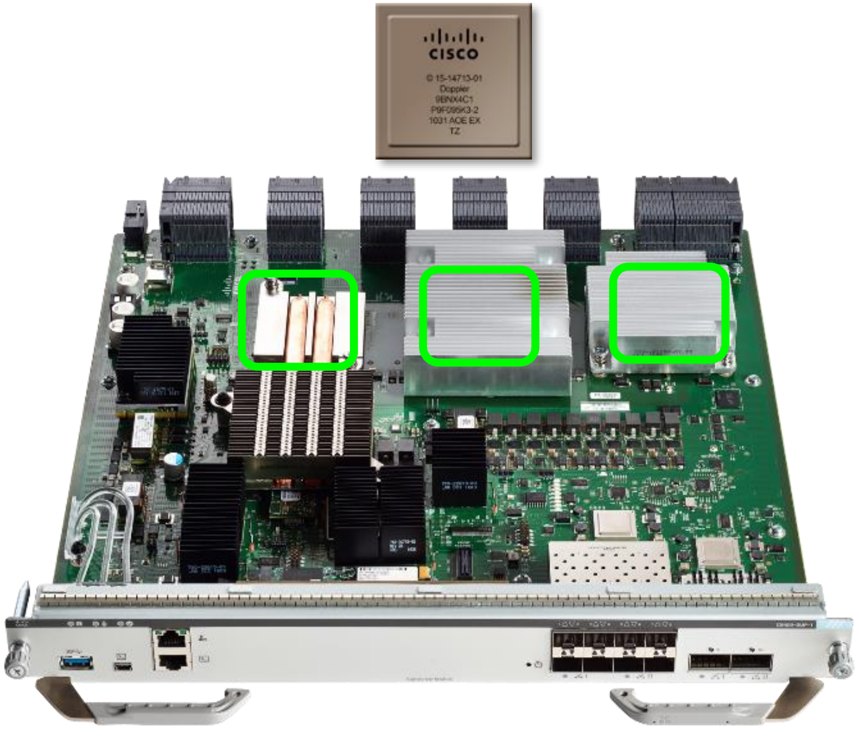
Up to 240GE Bandwidth

Up to 2X to 4X forwarding + TCAM

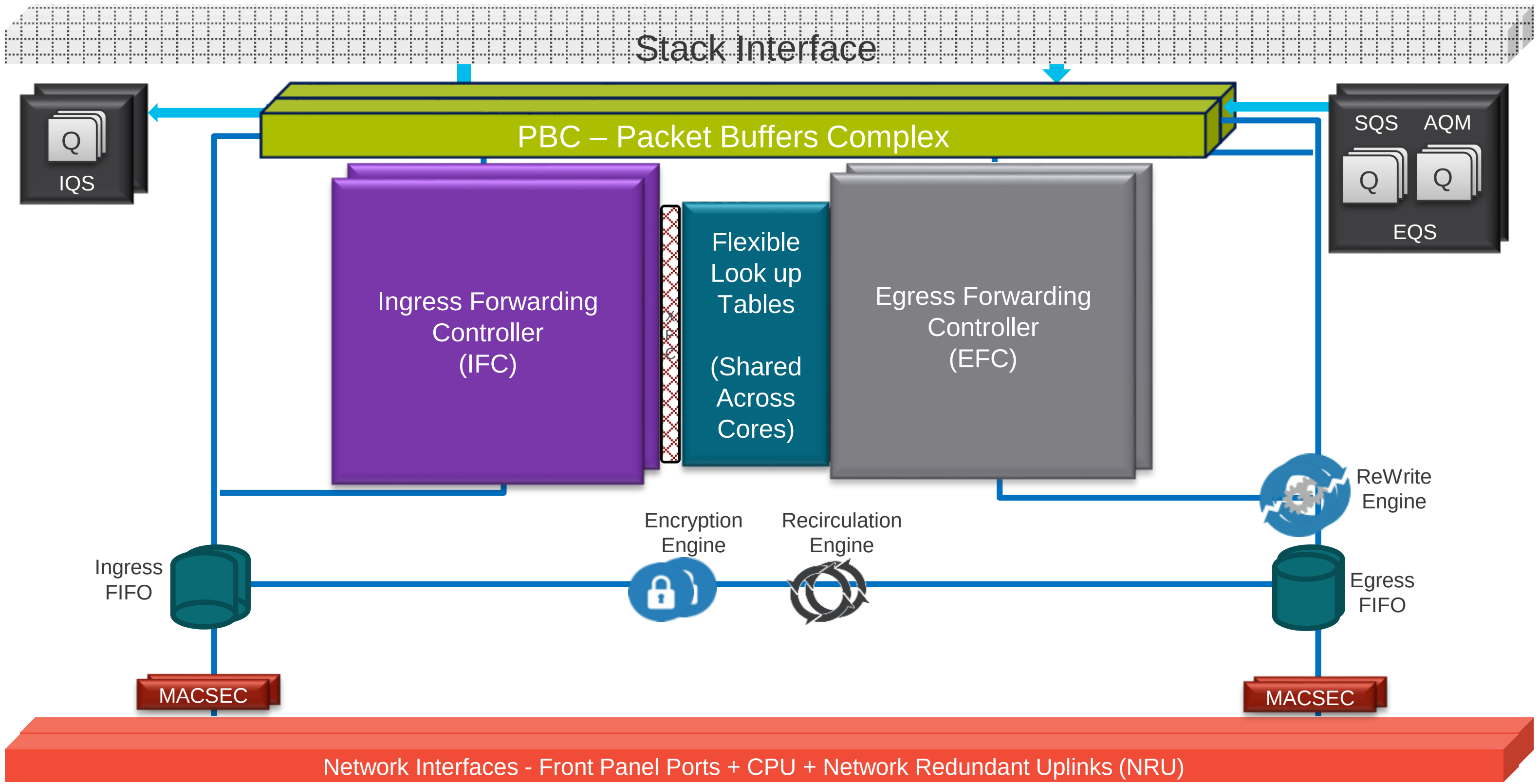
Embedded Microcontrollers

Up to 32MB Packet Buffer

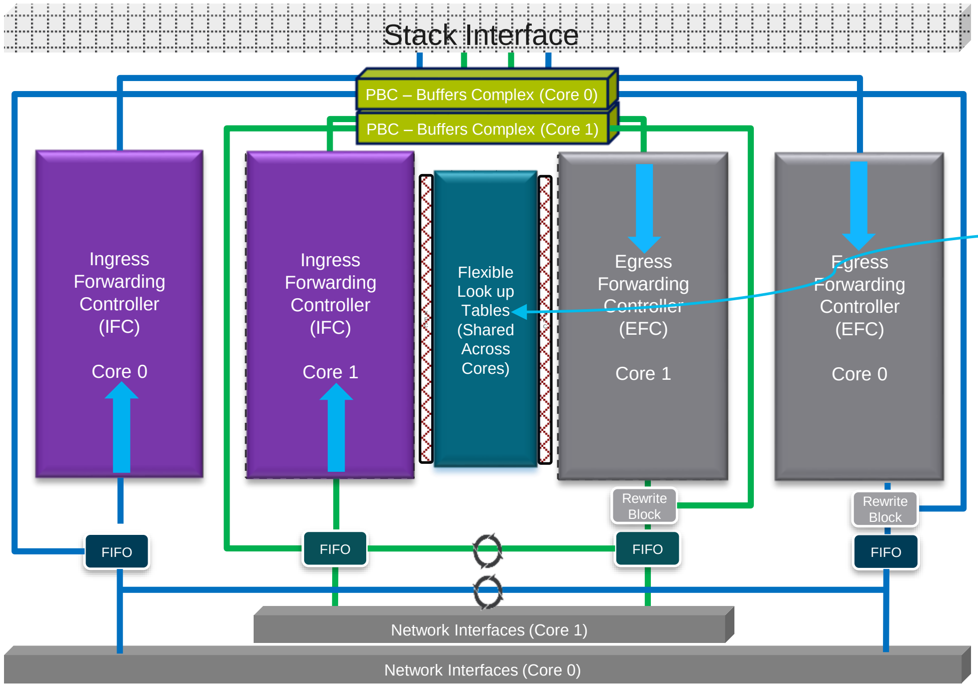
Up to 64K x2 Netflow Records



UADP 2.0 – 2 Cores



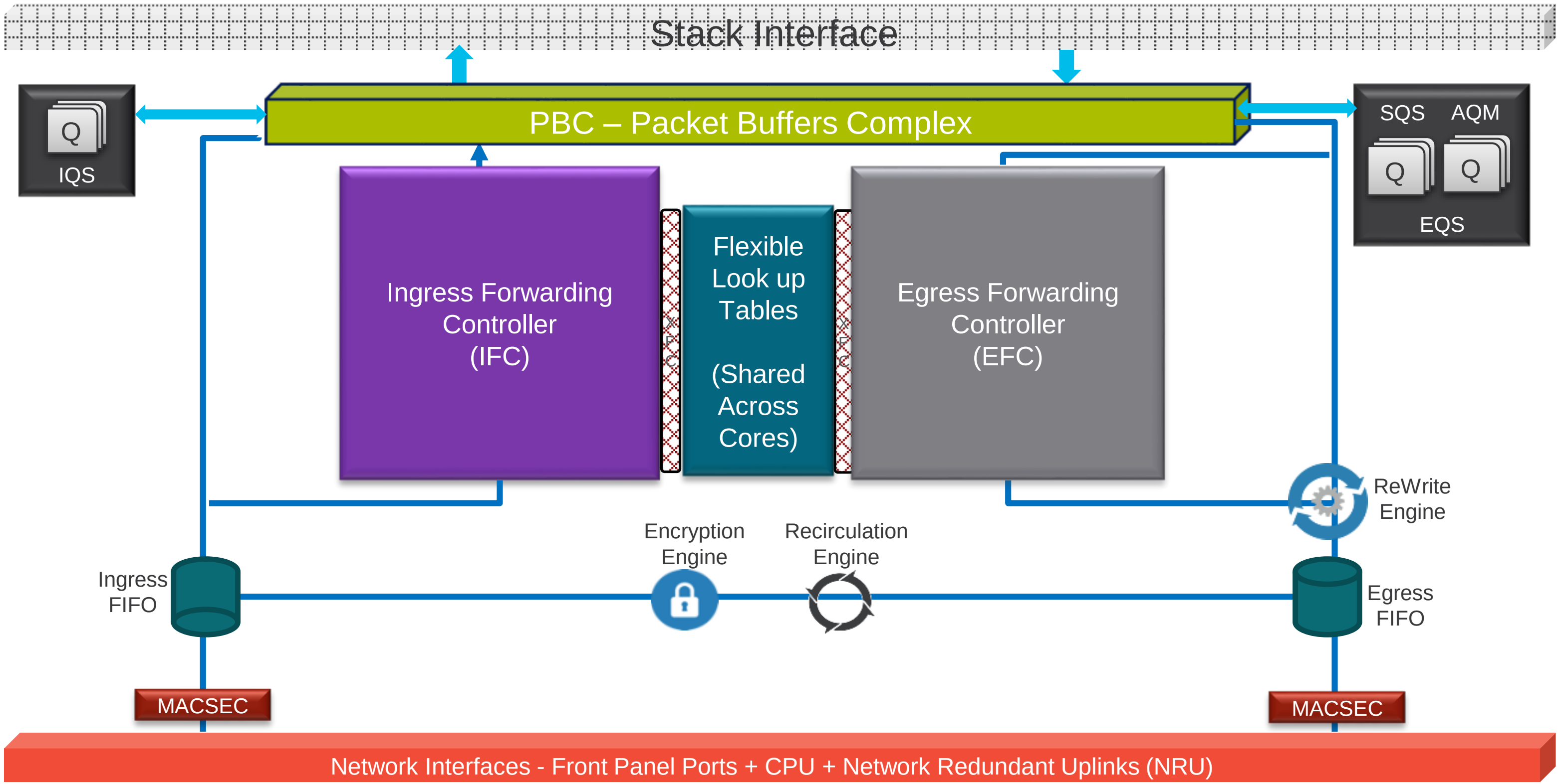
Shared Lookup Tables



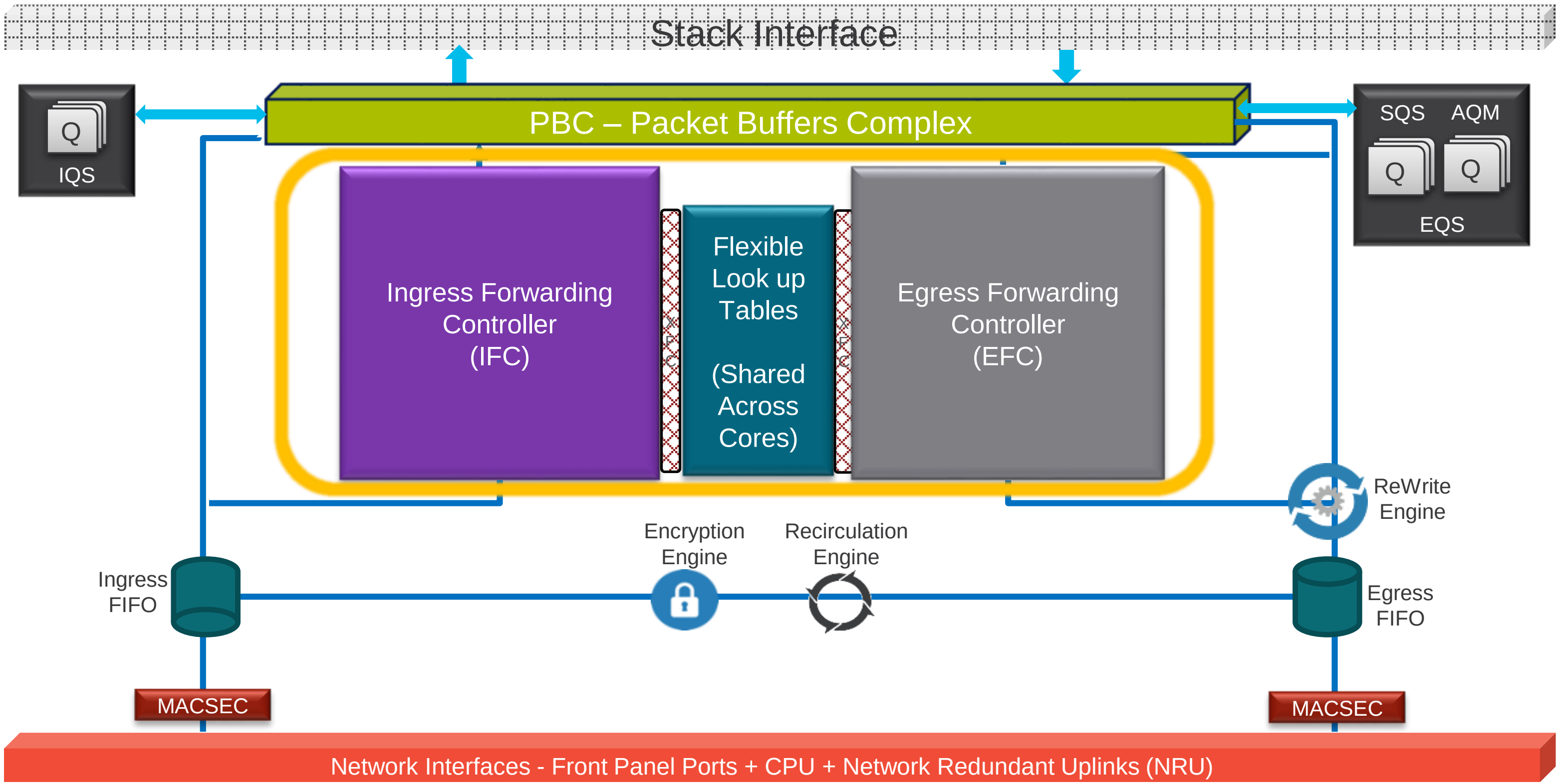
Lookup Tables shared across the cores

Scales better at High Performance

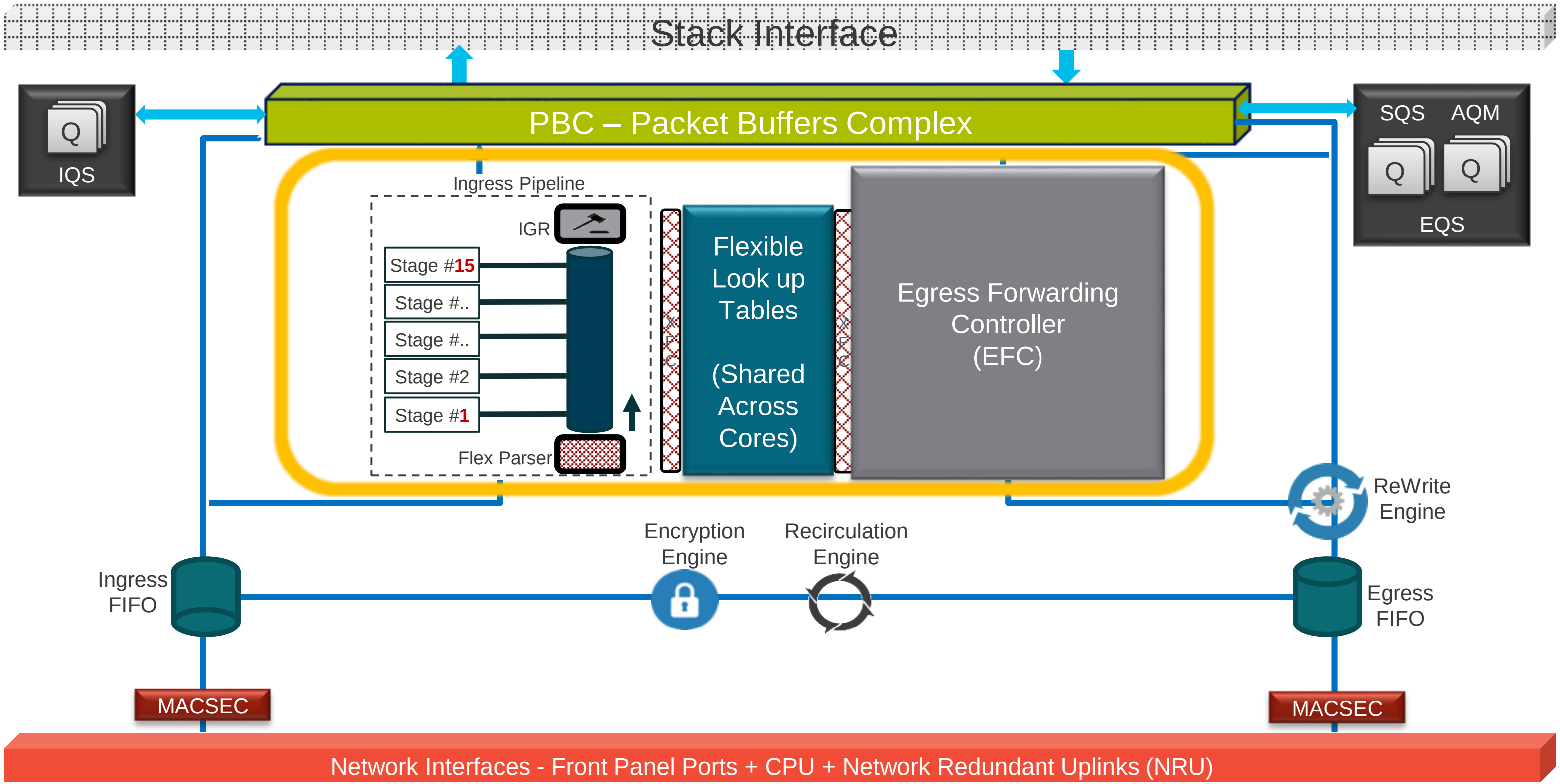
UADP 2.0 – Forwarding Pipelines



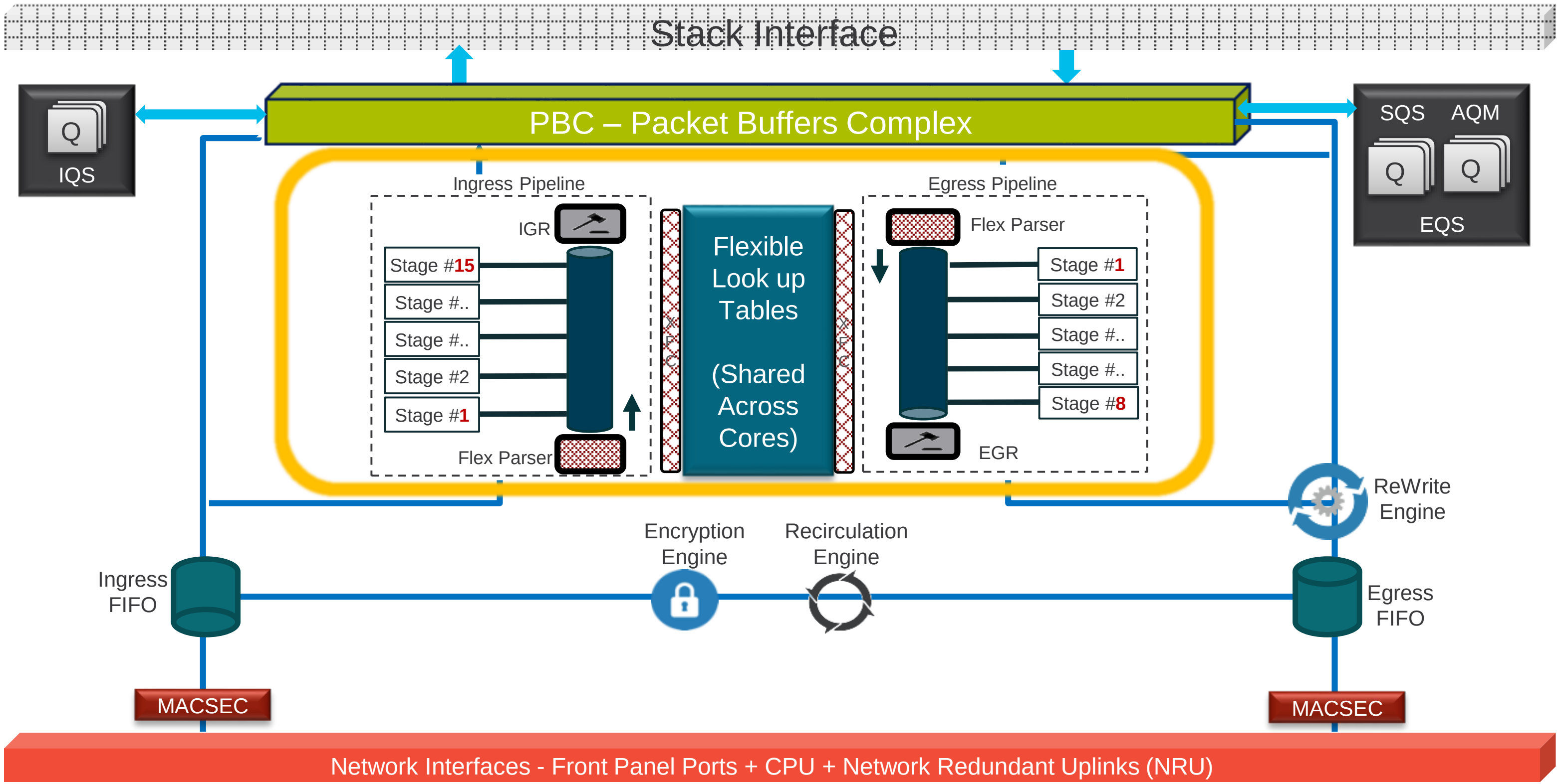
UADP 2.0 – Forwarding Pipelines



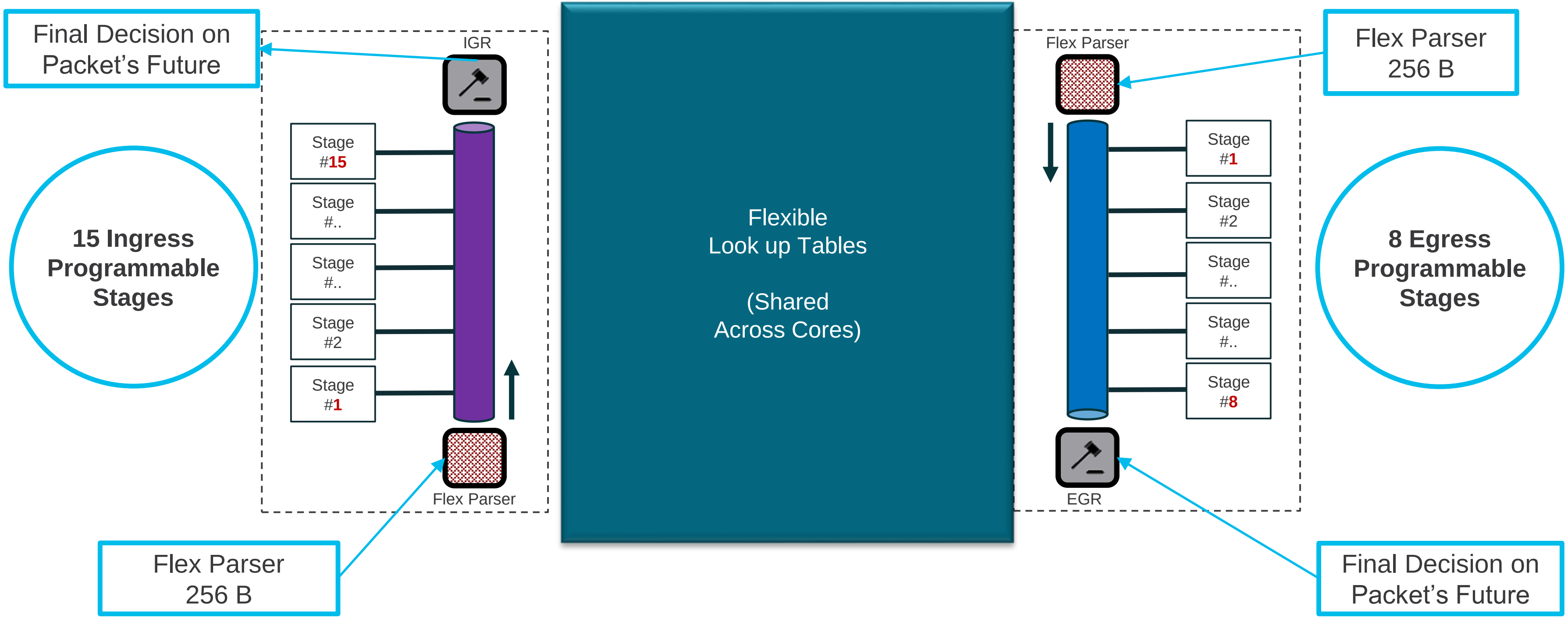
UADP 2.0 – Forwarding Pipelines



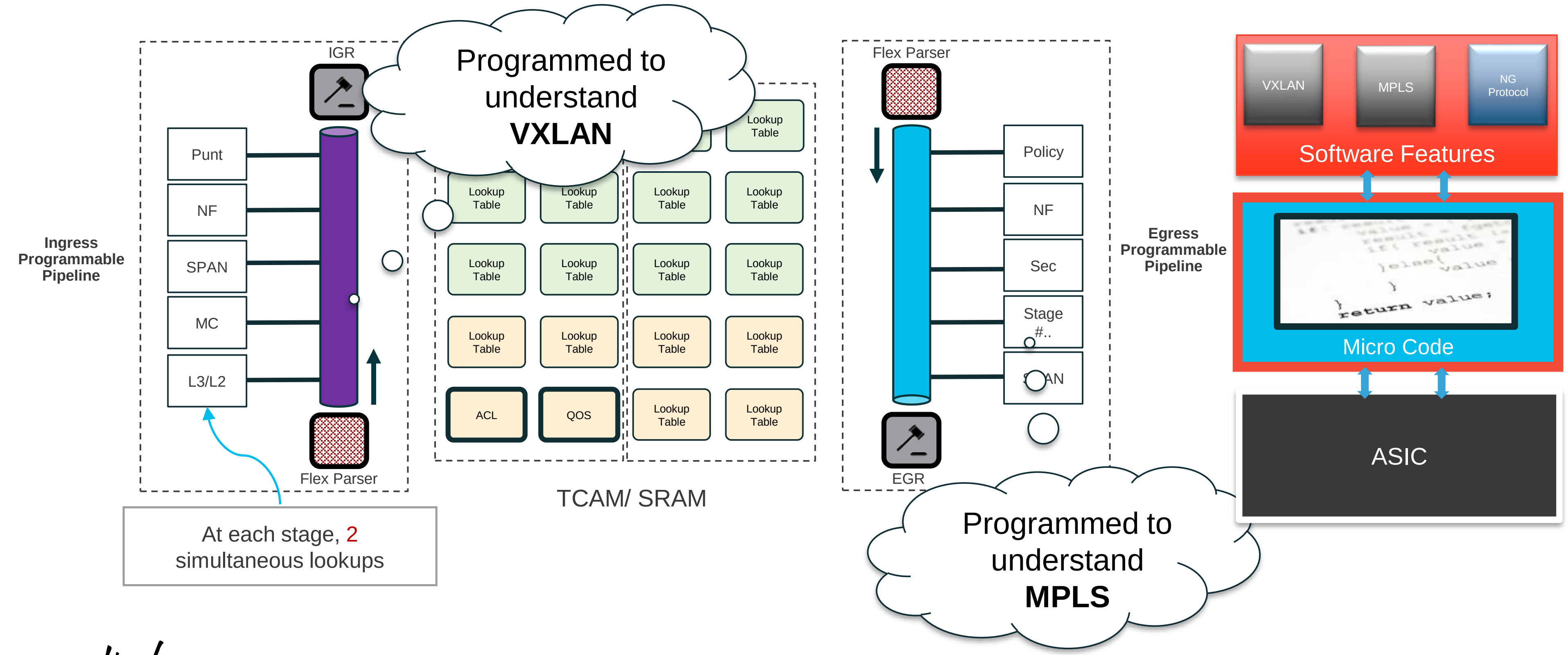
UADP 2.0 – Forwarding Pipelines



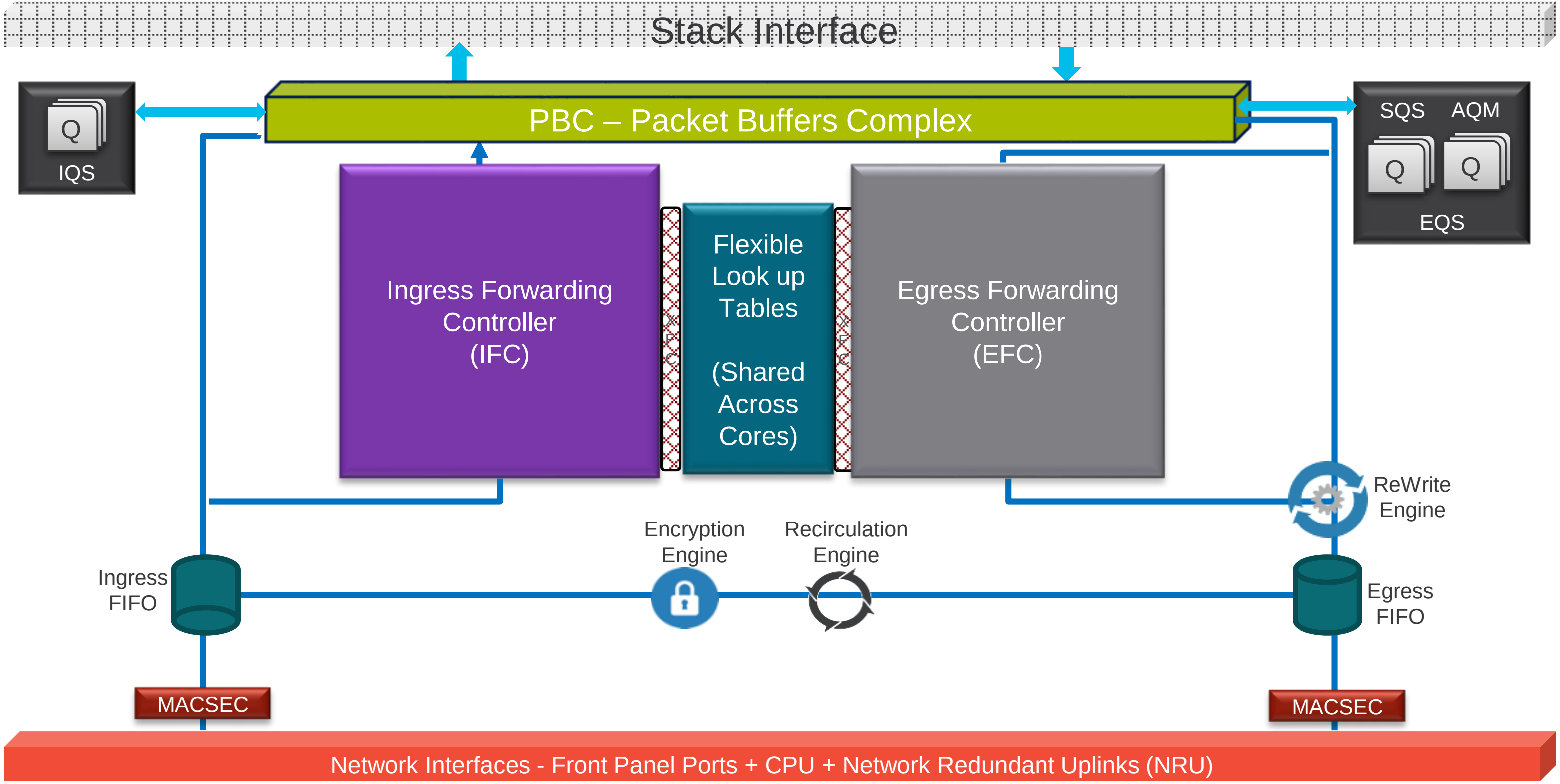
Forwarding - Under the covers



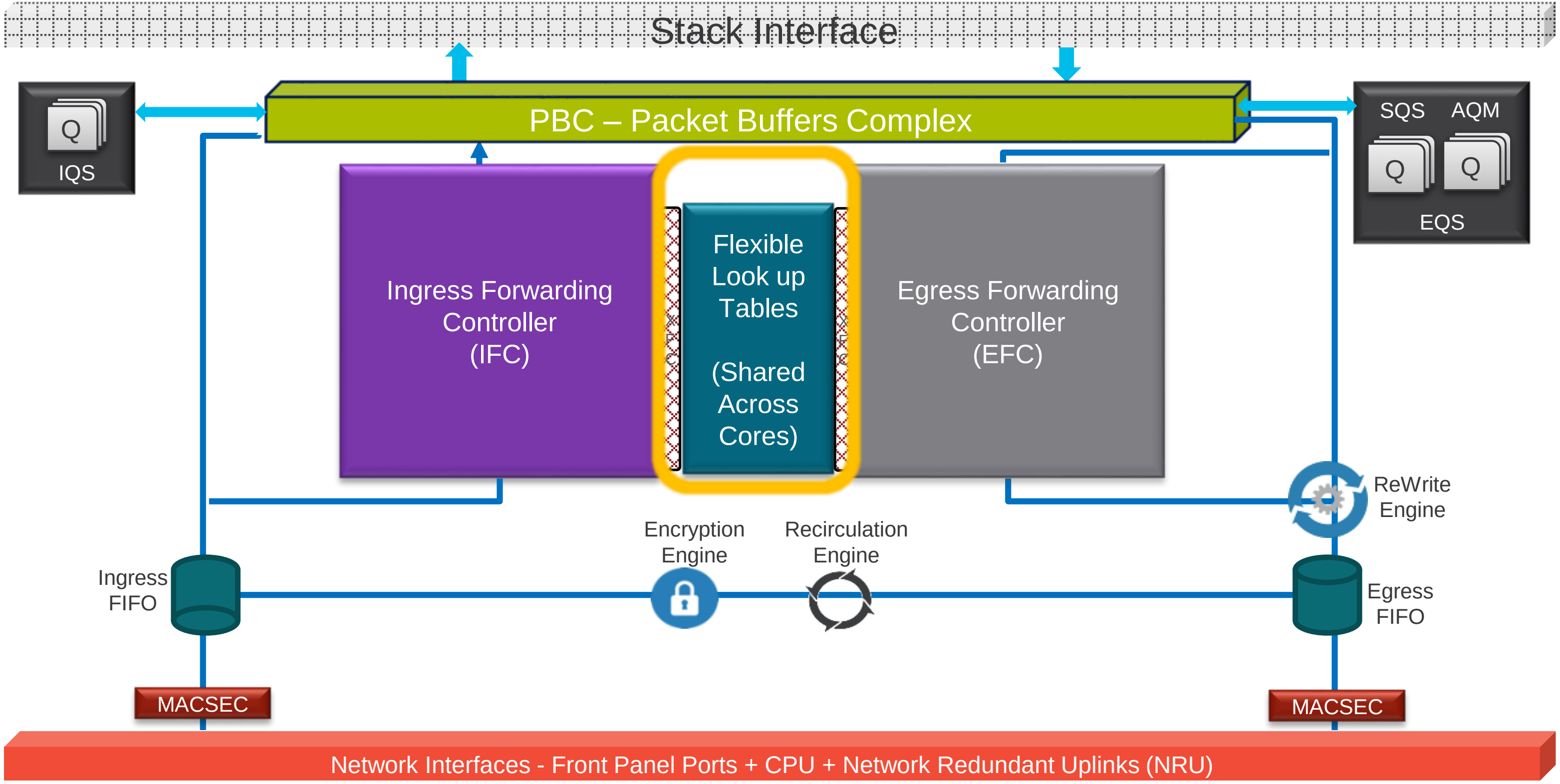
Microcode programs UADP 2.0



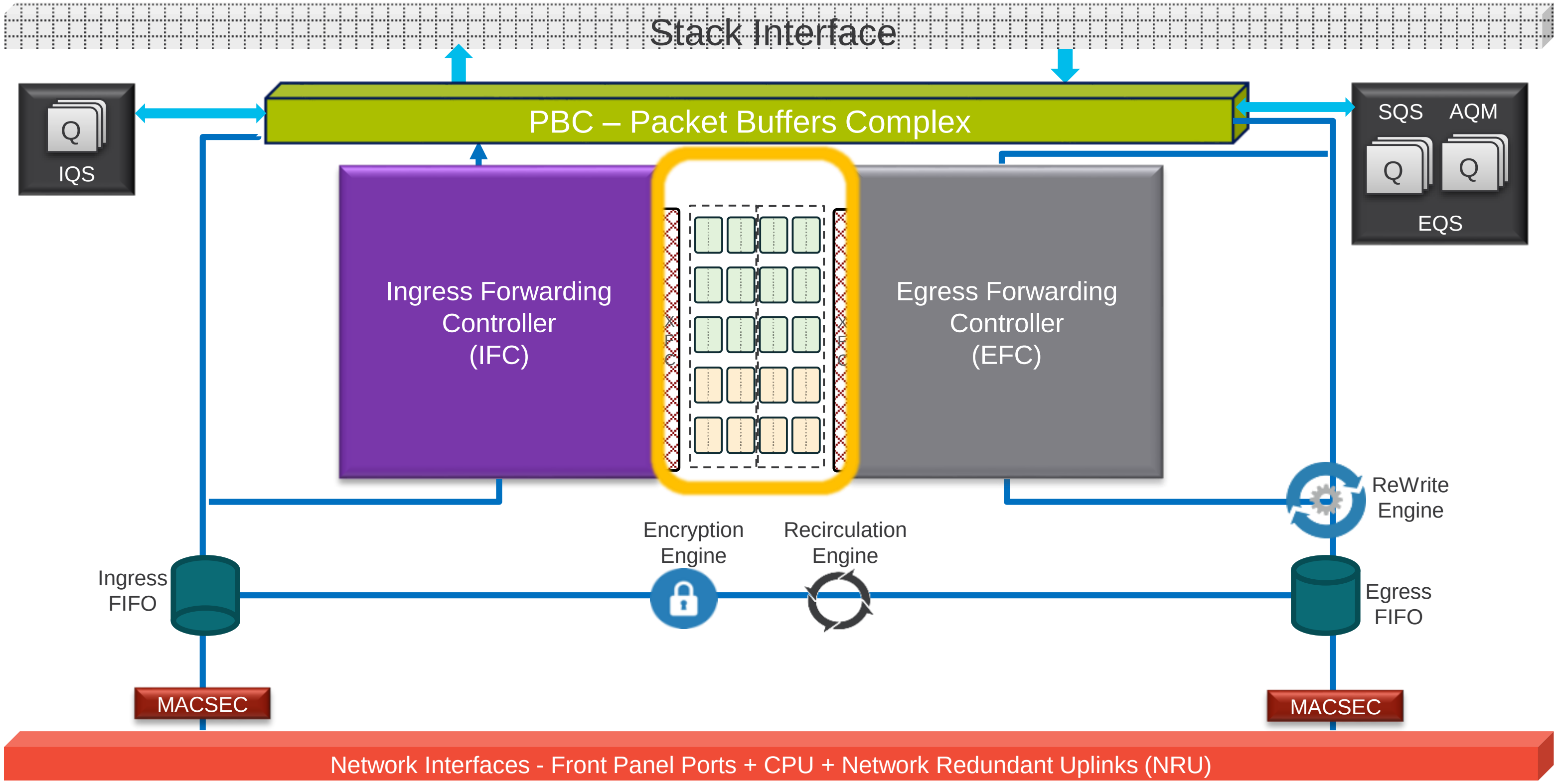
UADP 2.0 – Look up Tables



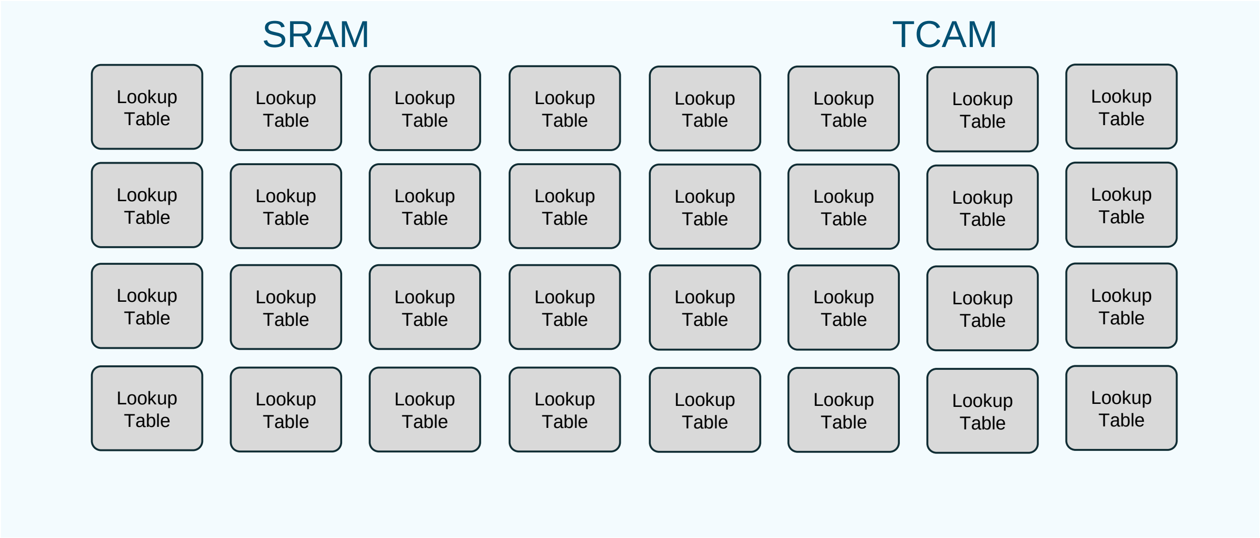
UADP 2.0 – Look up Tables



UADP 2.0 – Look up Tables



Flex Tables


SRAM
TCAM
Lookup Table
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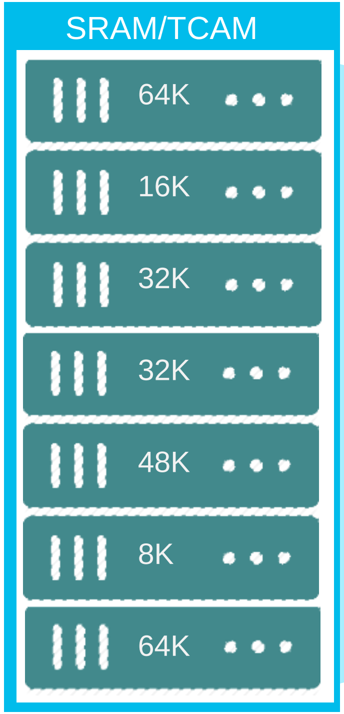
Lookup Table

Custom ASIC Templates for Universal Deployment

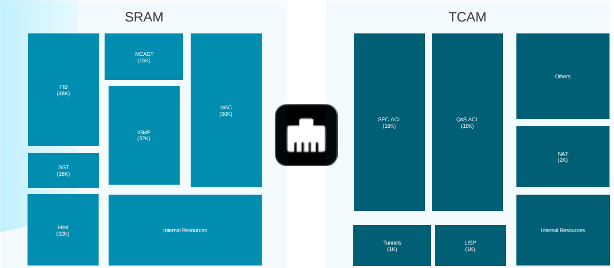


- MAC
- IPv4/IPv6
- VACL
- PACL
- RACL
- SGACL
- QoS
- NAT
- SPAN
- CoPP

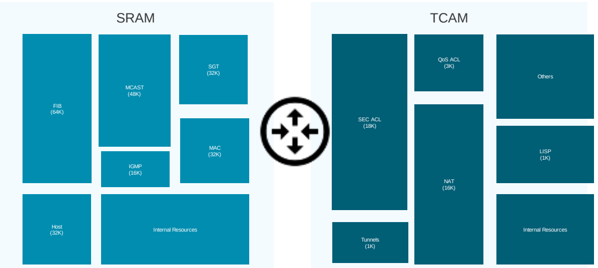
Flexible ASIC Templates

Access-Edge Template	Core-Border Template	Aggregation Template	Collapsed Core-WAN Template
Client Scale	L3 & Cross Domain Policy	Mix of L2/L3 Capabilities	L3 & NAT

Customize table size for each function based on the place in the network

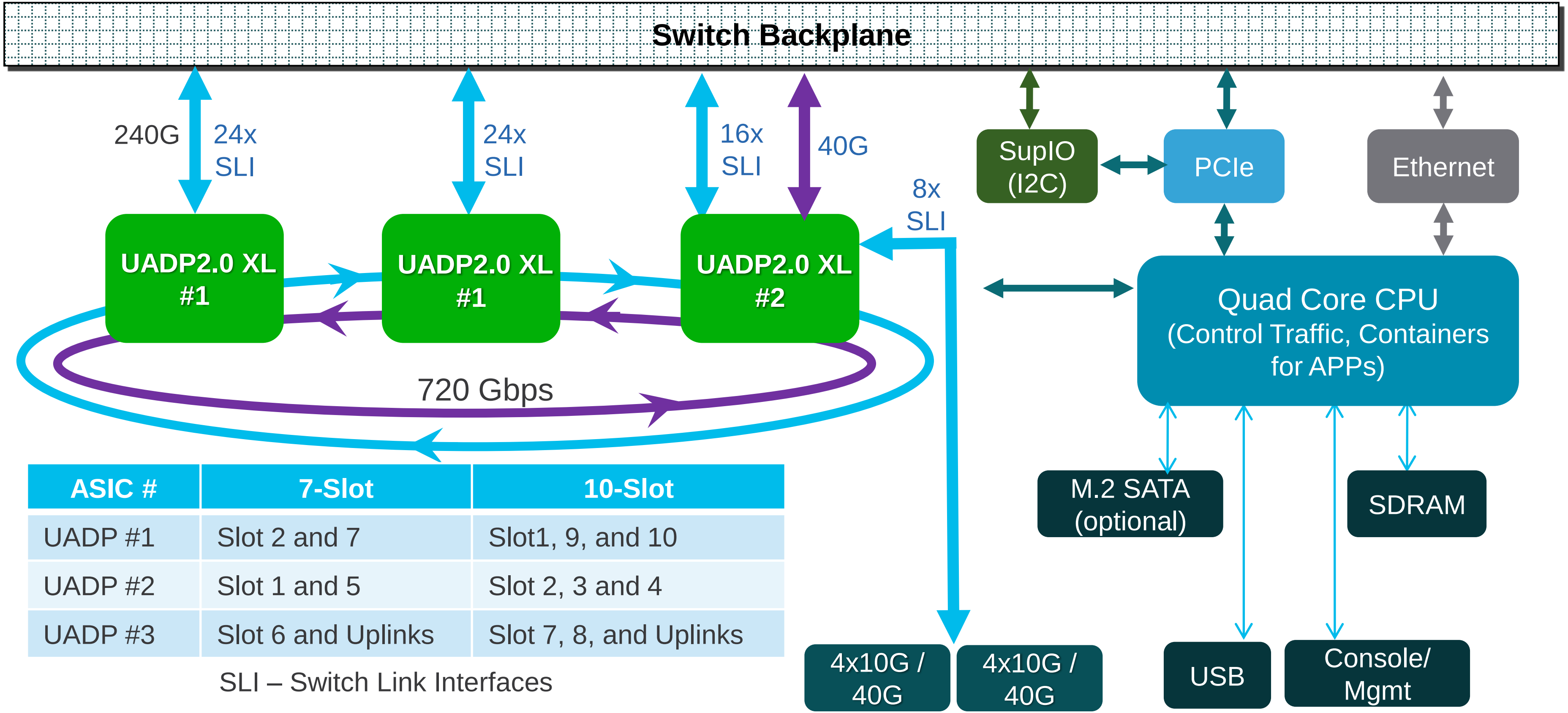


Access-Edge Template



Core-Border Template

Sup-1/Sup-1XL Block Diagram



ASIC #	7-Slot	10-Slot
UADP #1	Slot 2 and 7	Slot1, 9, and 10
UADP #2	Slot 1 and 5	Slot 2, 3 and 4
UADP #3	Slot 6 and Uplinks	Slot 7, 8, and Uplinks

SLI – Switch Link Interfaces

Port to ASIC Mapping

FYI

```
switch#show platform software fed active ifm mappings lpn
Mappings Table
LPN          ASIC  Port  Interface                    IF_ID              Active
-------------------------------------------------------------------------------
1            1     0     GigabitEthernet1/0/1         0x000000c6         Y
<SNIP>
48           1     47    GigabitEthernet1/0/48        0x00000036         Y
1            0     0     GigabitEthernet2/0/1         0x00000037         Y
<SNIP>
48           0     47    GigabitEthernet2/0/48        0x000000c5         Y
1            2     17    TenGigabitEthernet3/0/1      0x00000067         Y
<SNIP>
8            2     10    TenGigabitEthernet3/0/8      0x0000006e         Y
9            2     18    FortyGigabitEthernet3/0/9    0x0000006f         Y
10           2     19    FortyGigabitEthernet3/0/10   0x00000070         Y
1            2     3     TenGigabitEthernet4/0/1      0x00000071         Y
<SNIP>
8            2     4     TenGigabitEthernet4/0/8      0x00000078         Y
9            2     8     FortyGigabitEthernet4/0/9    0x00000079         Y
10           2     9     FortyGigabitEthernet4/0/10   0x0000007a         Y
1            1     0     TenGigabitEthernet5/0/1      0x0000007b         Y
<SNIP>
24           1     23    TenGigabitEthernet5/0/24     0x00000092         Y
1            2     24    GigabitEthernet6/0/1         0x00000093         Y
<SNIP>
24           2     69    GigabitEthernet6/0/24        0x000000aa         Y
25           2     30    TenGigabitEthernet6/0/25     0x000000ab         Y
<SNIP>
48           2     63    TenGigabitEthernet6/0/48     0x000000c2         Y

switch#
```

Catalyst 9407R

```
Mod Card Type
--- ------------------------------
1   48-Port UPOE 10/100/1000 (RJ-45)
2   48-Port 10/100/1000 (RJ-45)
3   Supervisor 1 Module
4   Supervisor 1 Module
5   24-Port 10 Gigabit Ethernet (SFP+)
6   48-Port UPOE w/ 24p mGig 24p RJ-45
```

Sup-1/Sup-1XL Uplink - Single Sup

Sup-1/Sup-1XL Uplink - Single Sup

Active

Disabled

Sup-1/Sup-1XL Uplink - Single Sup

Active

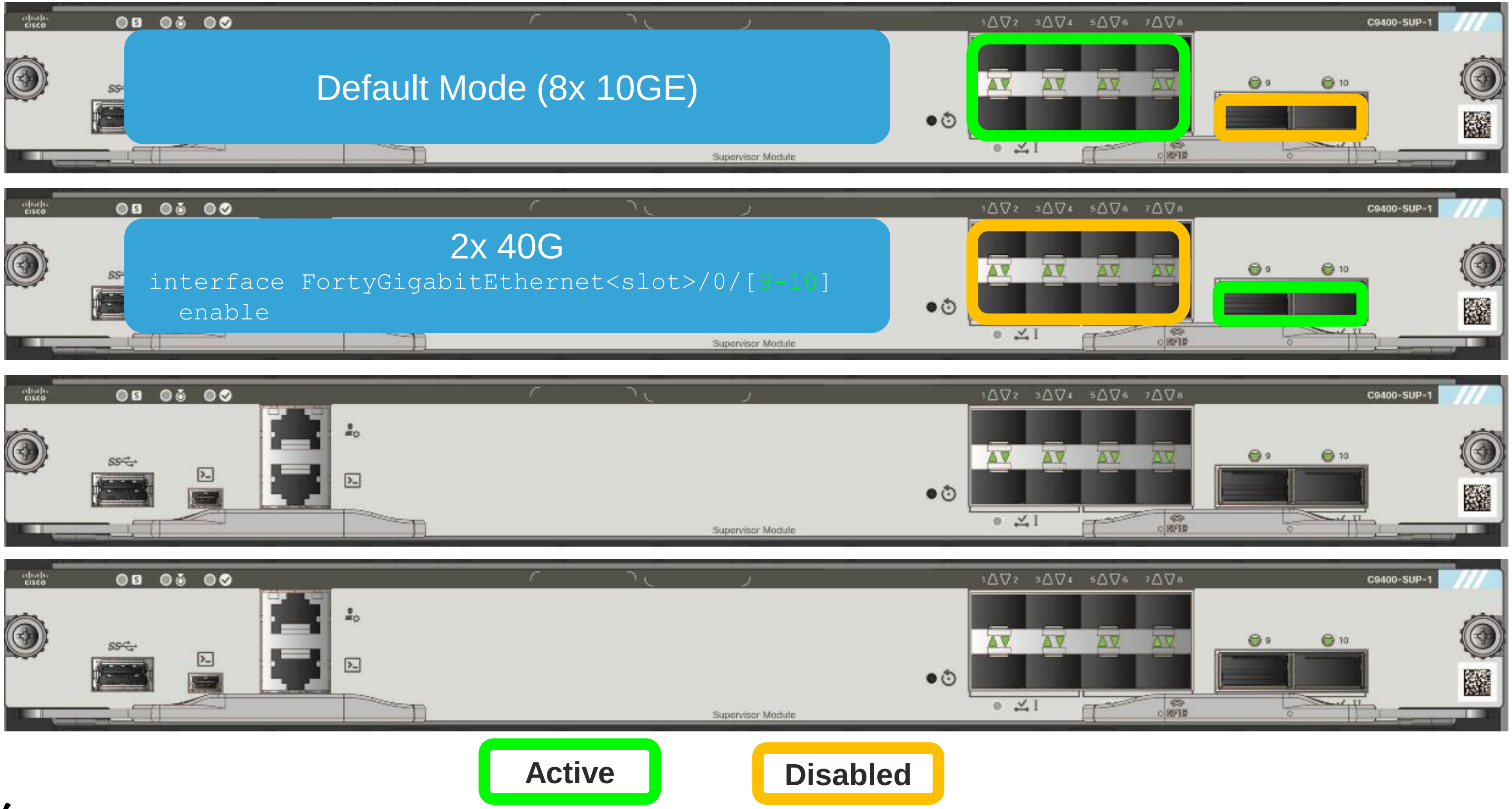
Disabled

Sup-1/Sup-1XL Uplink - Single Sup

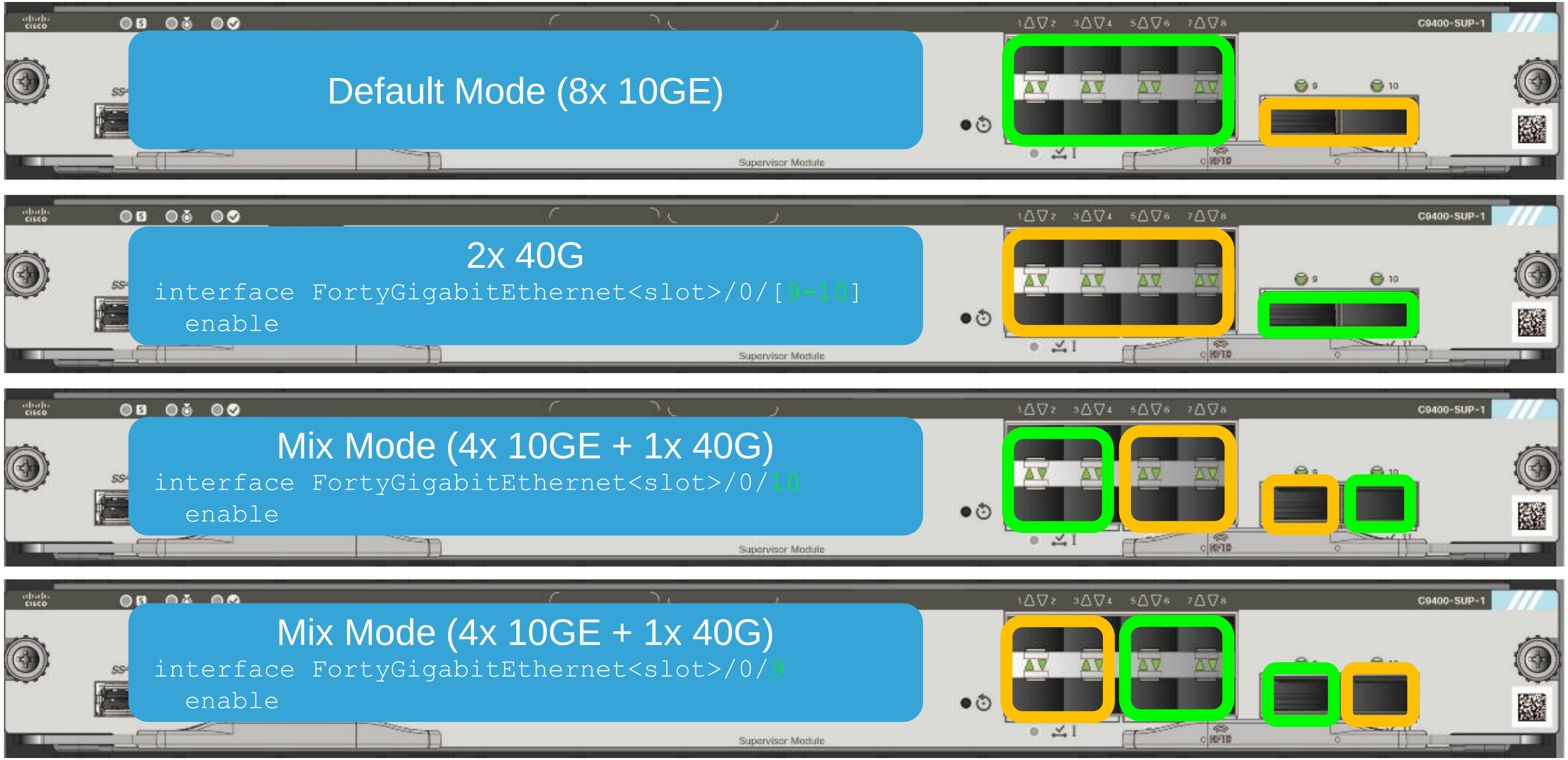
Active

Disabled

Sup-1/Sup-1XL Uplink - Single Sup

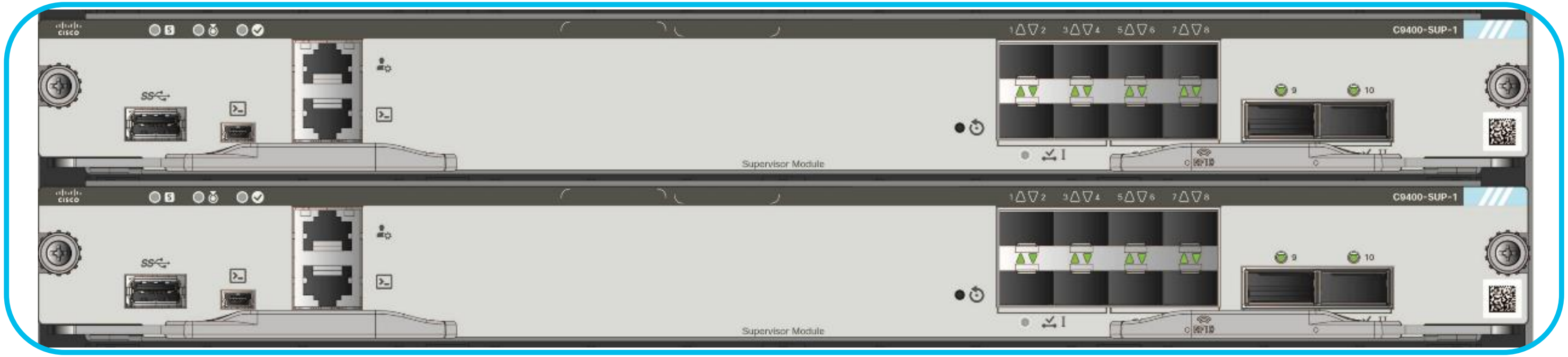


Sup-1/Sup-1XL Uplink - Single Sup

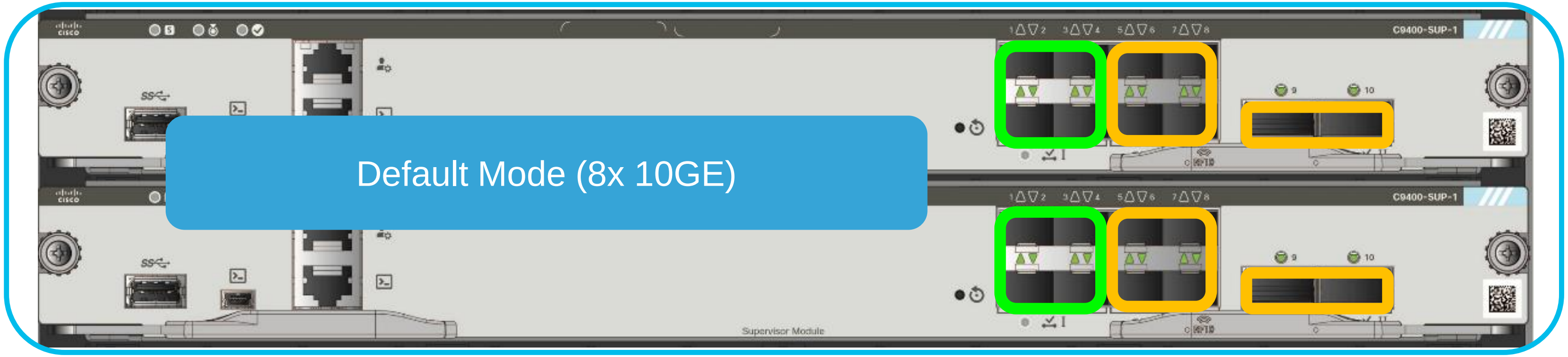


Active **Disabled**

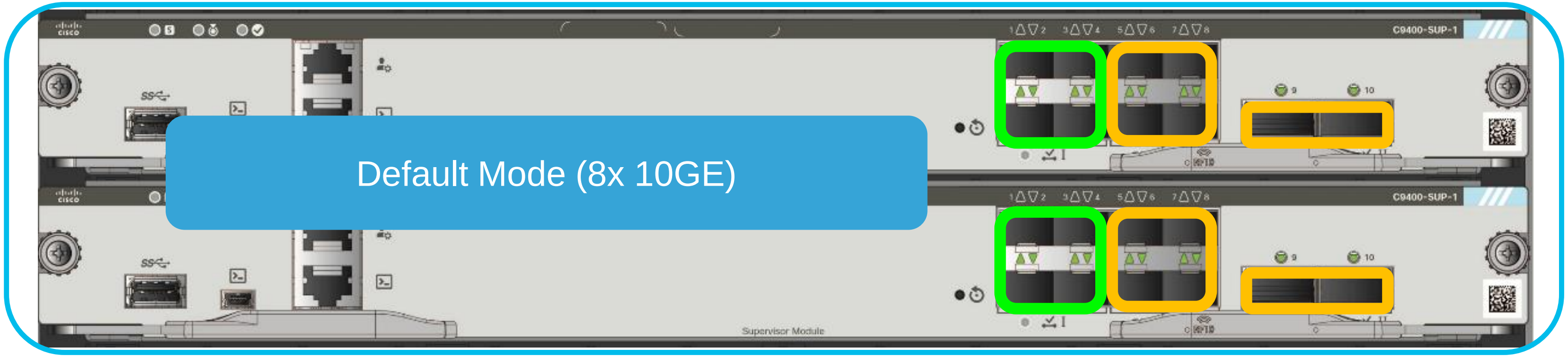
Sup-1/Sup-1XL Dual Sups - Uplink Redundancy



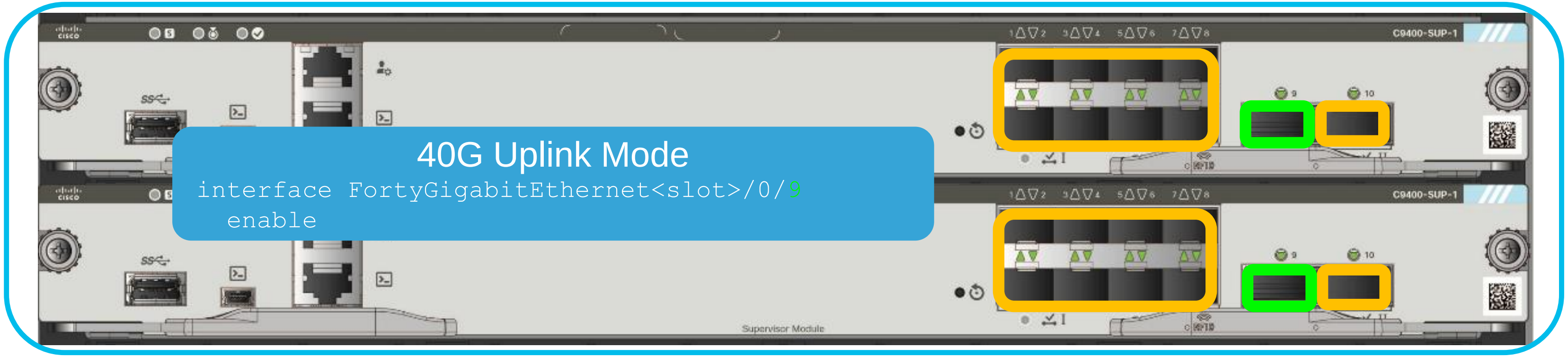
Sup-1/Sup-1XL Dual Sups - Uplink Redundancy



Sup-1/Sup-1XL Dual Sups - Uplink Redundancy



Default Mode (8x 10GE)

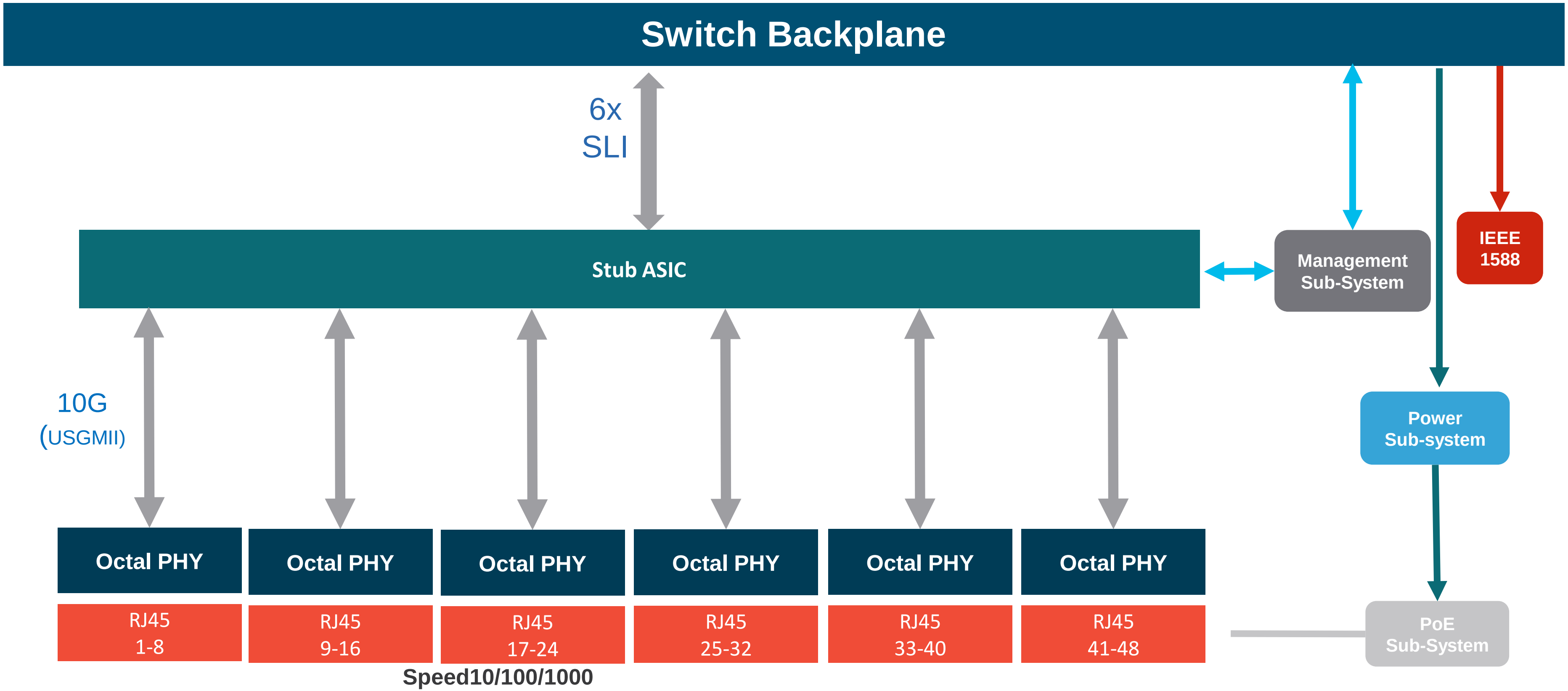


40G Uplink Mode

```
interface FortyGigabitEthernet<slot>/0/9
  enable
```

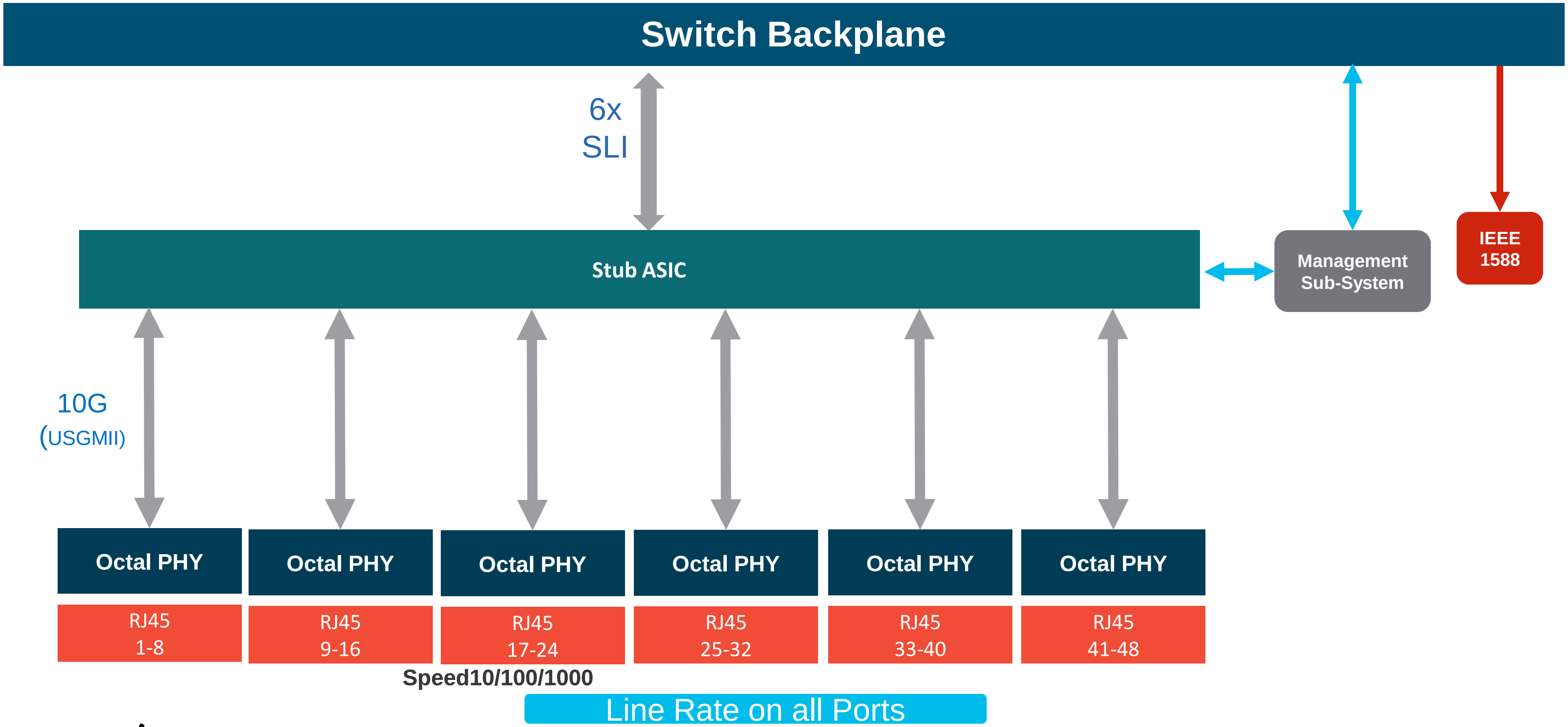
Active **Disabled**

48x1G RJ45 Line Card (UPoE)

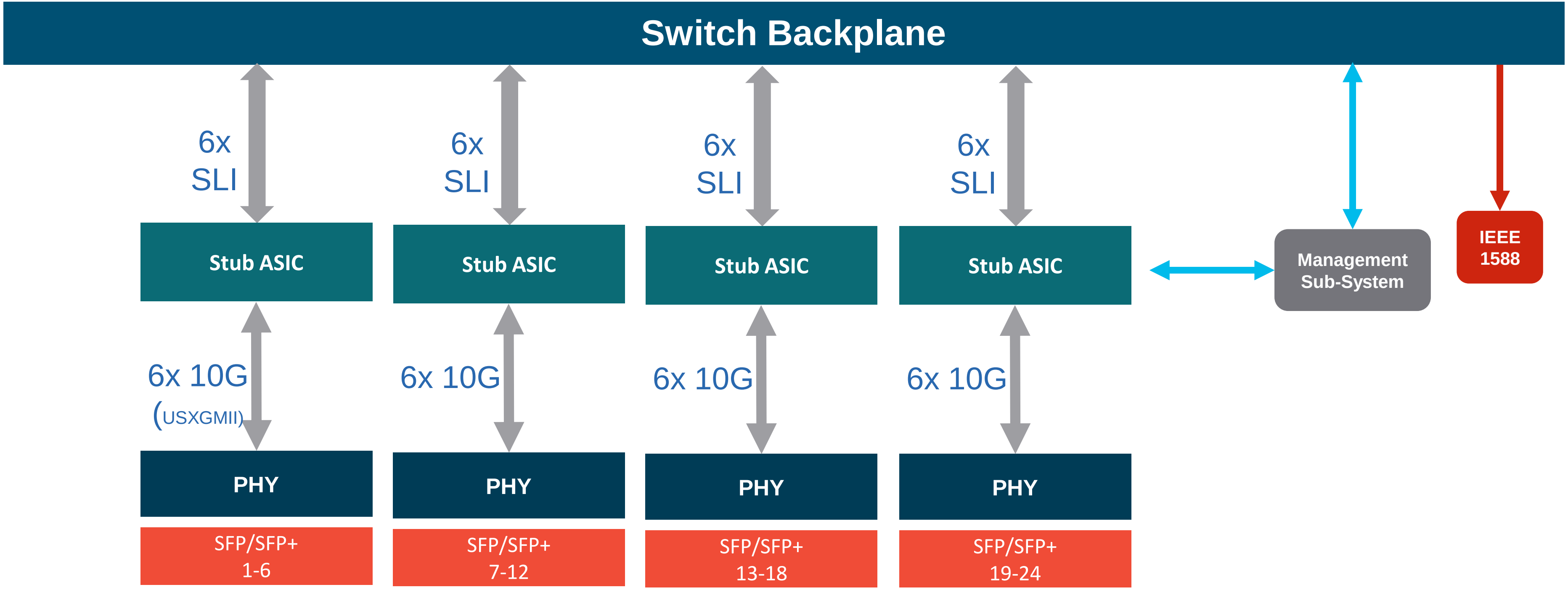


Line Rate on all Ports; UPoE on all Ports

48x1G Line Card (RJ45 Data or SFP)



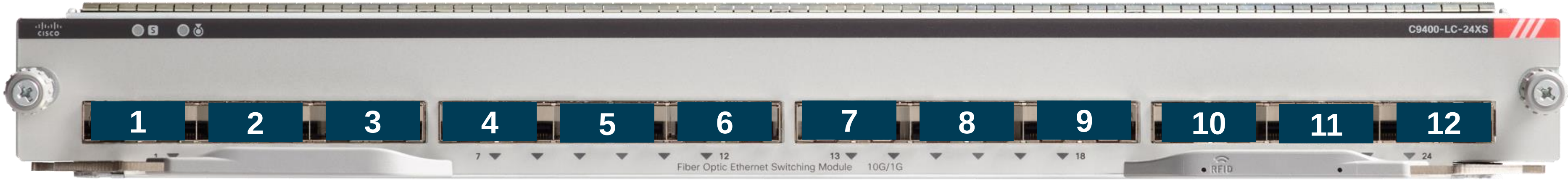
24x 1/10G SFP/SFP+ Line Card



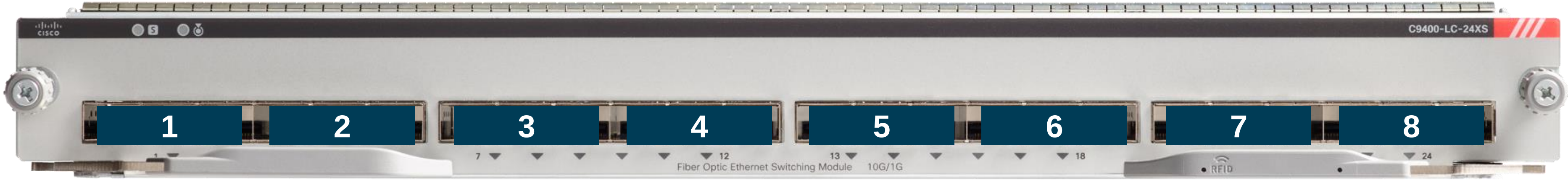
Sup-1: 80G with C9410R and C9407R
Sup-1XL: 80G with C9410R; 120G with C9407R

C9400-LC-24XS Port-Group

7 Slot Chassis: 12 Port-Group



10 Slot Chassis: 8 Port-Group



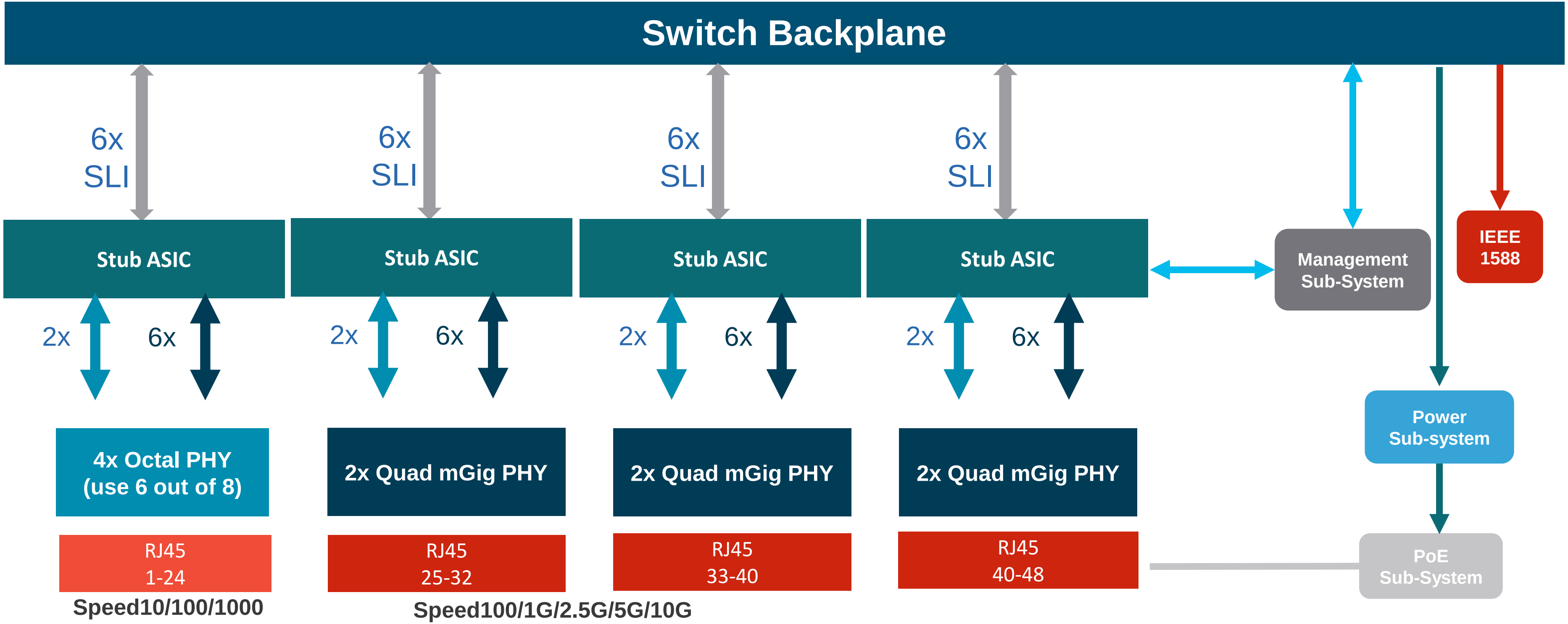
FYI

C9400-LC-24XS Port-Group – 7 Slot Chassis

- Bandwidth shared within port-group
- 12 port-group in the 7 slot chassis
- For 10G line rate performance:
  - Configure: “`hw-module subslot <slot#/0> mode dynamic`”
  - 8 Port with C9400-Sup-1
  - 12 port with C9400-Sup-1XL

```
R4-C94-2041#show platform hardware iomd 5/0 portgroups
Port  Interface                       Status    Interface  Group
Max
Group                                            Bandwith   Bandwidth

1     TenGigabitEthernet5/0/1          up         10G
1     TenGigabitEthernet5/0/2          down       10G        10G

2     TenGigabitEthernet5/0/3          up         10G
2     TenGigabitEthernet5/0/4          down       10G        10G

3     TenGigabitEthernet5/0/5          up         10G
3     TenGigabitEthernet5/0/6          down       10G        10G

4     TenGigabitEthernet5/0/7          up         10G
4     TenGigabitEthernet5/0/8          down       10G        10G
<SNIP>
11    TenGigabitEthernet5/0/21         up         10G
11    TenGigabitEthernet5/0/22         down       10G        10G


12    TenGigabitEthernet5/0/23         up         10G
12    TenGigabitEthernet5/0/24         down       10G        10G

R4-C94-2041#show
```

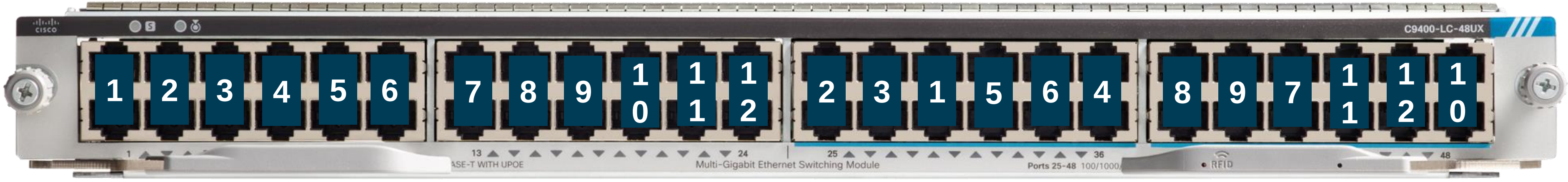
mGig RJ45 Line Card



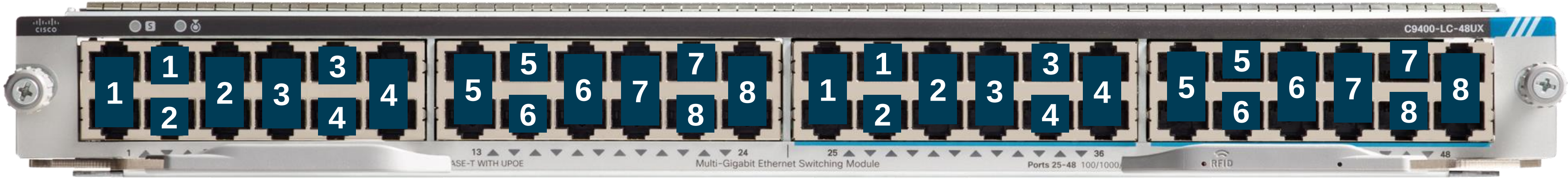
Sup-1: 80G with C9410R and C9407R
Sup-1XL: 80G with C9410R; 120G with C9407R

C9400-LC-48UX Port-Group

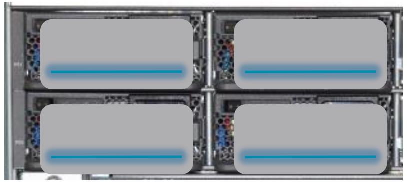
7 Slot Chassis: 12 Port-Group



10 Slot Chassis: 8 Port-Group



Power

	Normal	PS failure
Combined (Default)	 Load sharing on all PSs	 Load sharing on functional PSs
Redundant	 Load sharing on active PSs Standby PS in output disabled	 Standby PS becomes active System enters alarm state

Failed PS

ACTIVE

STANDBY

Power Redundancy: N+N and N+1

- Default active is PS1-4 and standby is PS5-8
- Standby power slots are configurable

```
SW(config)#power redundancy-mode redundant ?
  N+N   Redundant N+N (N is active, N is standby)
  N+1   Redundant N+N (N is active, 1 is standby)
SW(config)#power redundancy-mode redundant N+1 ?
  <1-8>   standby slot in N+N mode
SWR(config)#
```

- Default active is PS1-7 and standby is PS8
- Standby power slot is configurable

```
SW(config)#power redundancy-mode redundant ?
  N+N   Redundant N+N (N is active, N is standby)
  N+1   Redundant N+N (N is active, 1 is standby)
SW(config)#power redundancy-mode redundant N+1 ?
  <1-8>   standby slot in N+1 mode
SWR(config)#
```

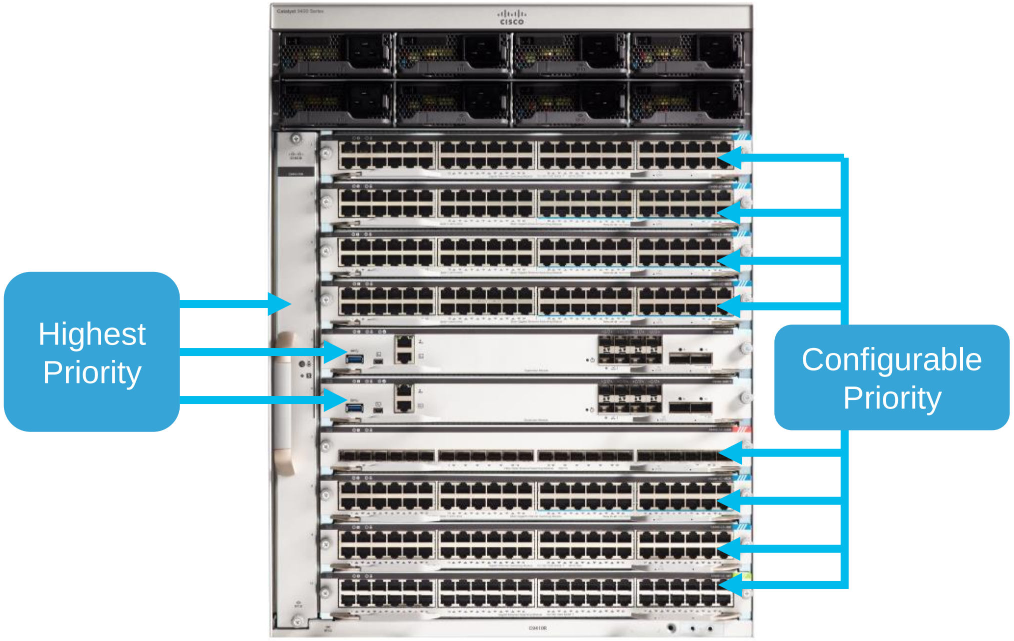
ACTIVE

STANDBY

Power Priority

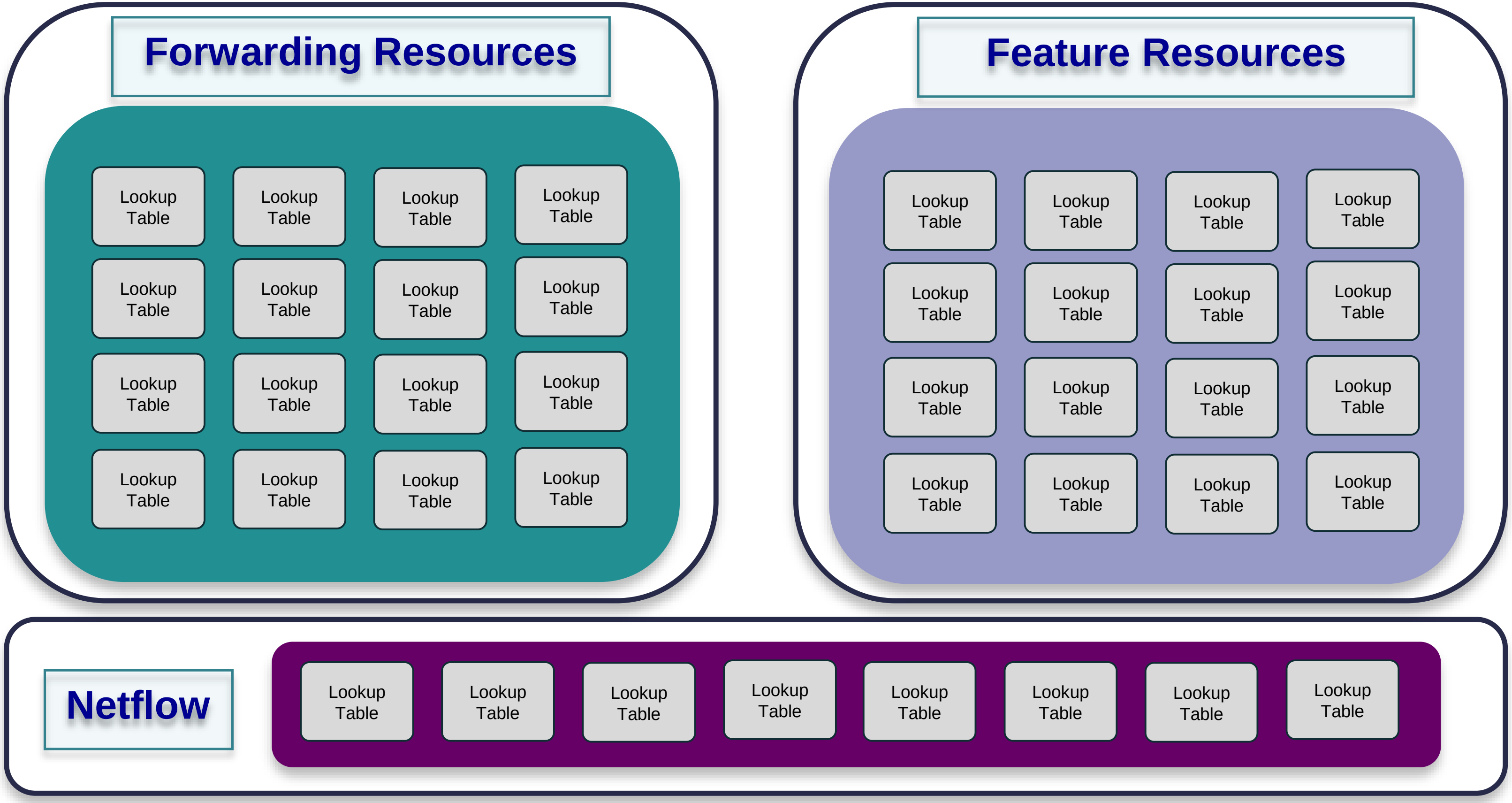
- All components in the system are assigned with power priority level
- Supervisors and Fan Tray has the same highest priority level
- Lower slot# has the higher power priority level by default if “power supply autoLC shutdown” is configured
- Configurable power priority for line card slots

```
C94(config)#power supply autoLC priority ?
  <1-7>  Physical slot number
  <cr>
C94(config)
```



Forwarding

Flex Tables


Forwarding Resources
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Feature Resources
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Netflow
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table
Lookup Table

Flex Tables

Forwarding Resources

- MAC
- Host Route
- IGMP Groups
- LPM Route
- Multicast Route
- SGT

Feature Resources

- Security ACL
- QoS ACL
- Service ACL
  - PBR/NAT
  - Netflow ACL
  - SPAN
  - MACsec
  - CoPP
  - Tunnel
  - LISP

Netflow

Netflow Entries

Flex Tables

Forwarding Resources

- MAC: 64K
- Host Route: 48K – 112K
- IGMP Groups: 16K
- LPM Route: 64K
- Multicast Route: 16K
- SGT: 16K

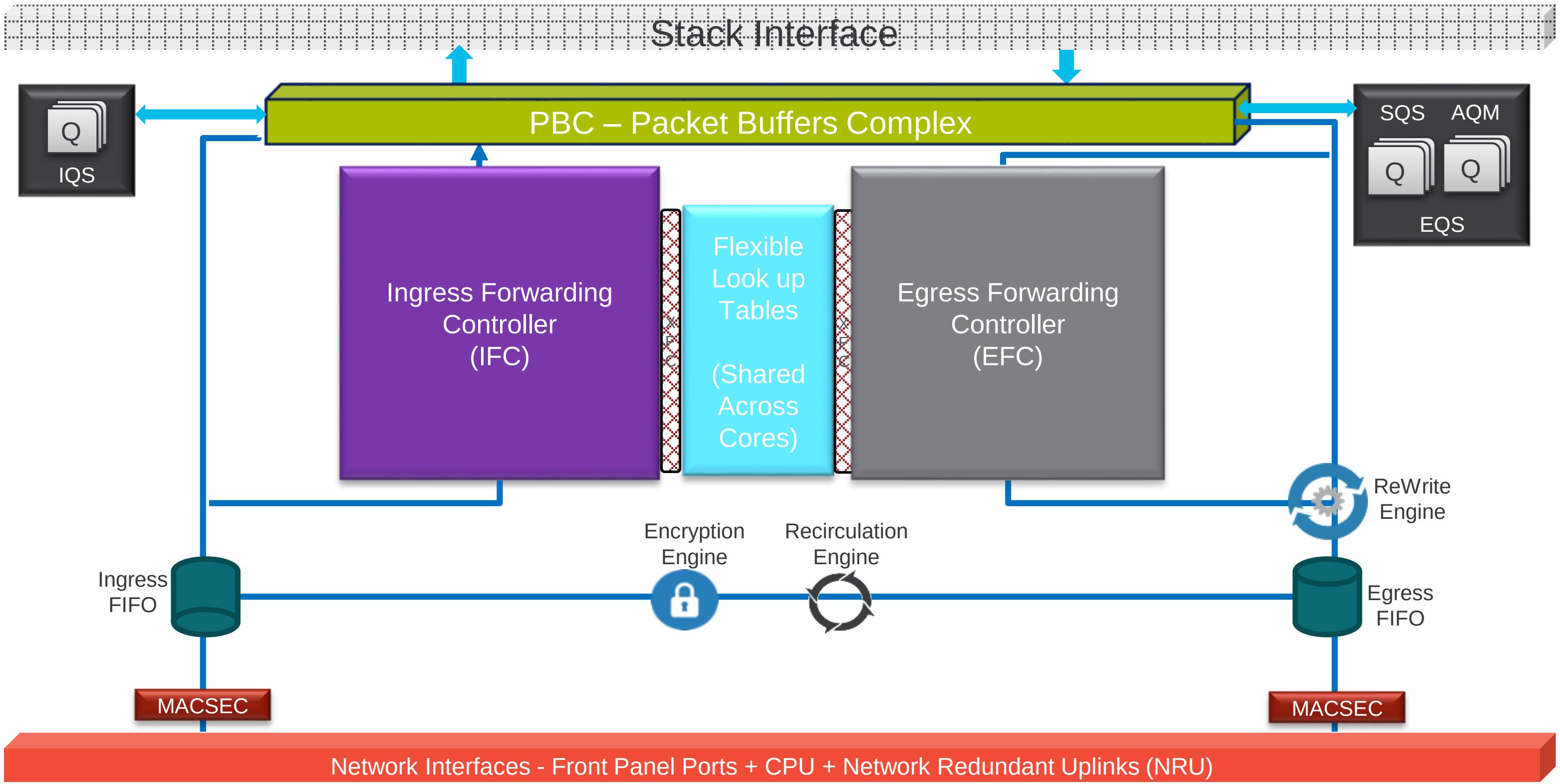
Feature Resources

- Security ACL: 18K
- QoS ACL: 18K
- Service ACL: 18K
  - PBR/NAT
  - Netflow ACL
  - SPAN
  - MACsec
  - CoPP
  - Tunnel
  - LISP

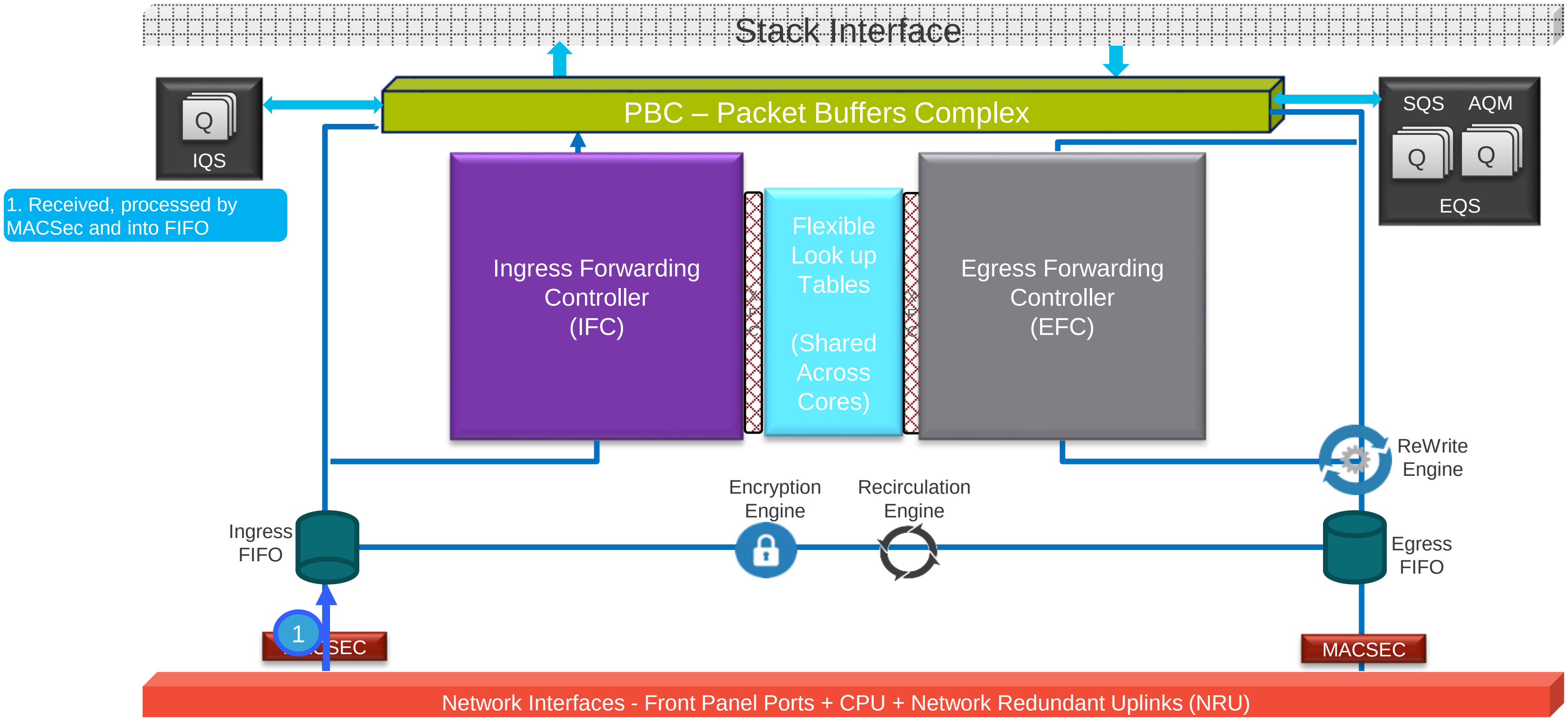
Netflow

Netflow Entries: 128K per ASIC

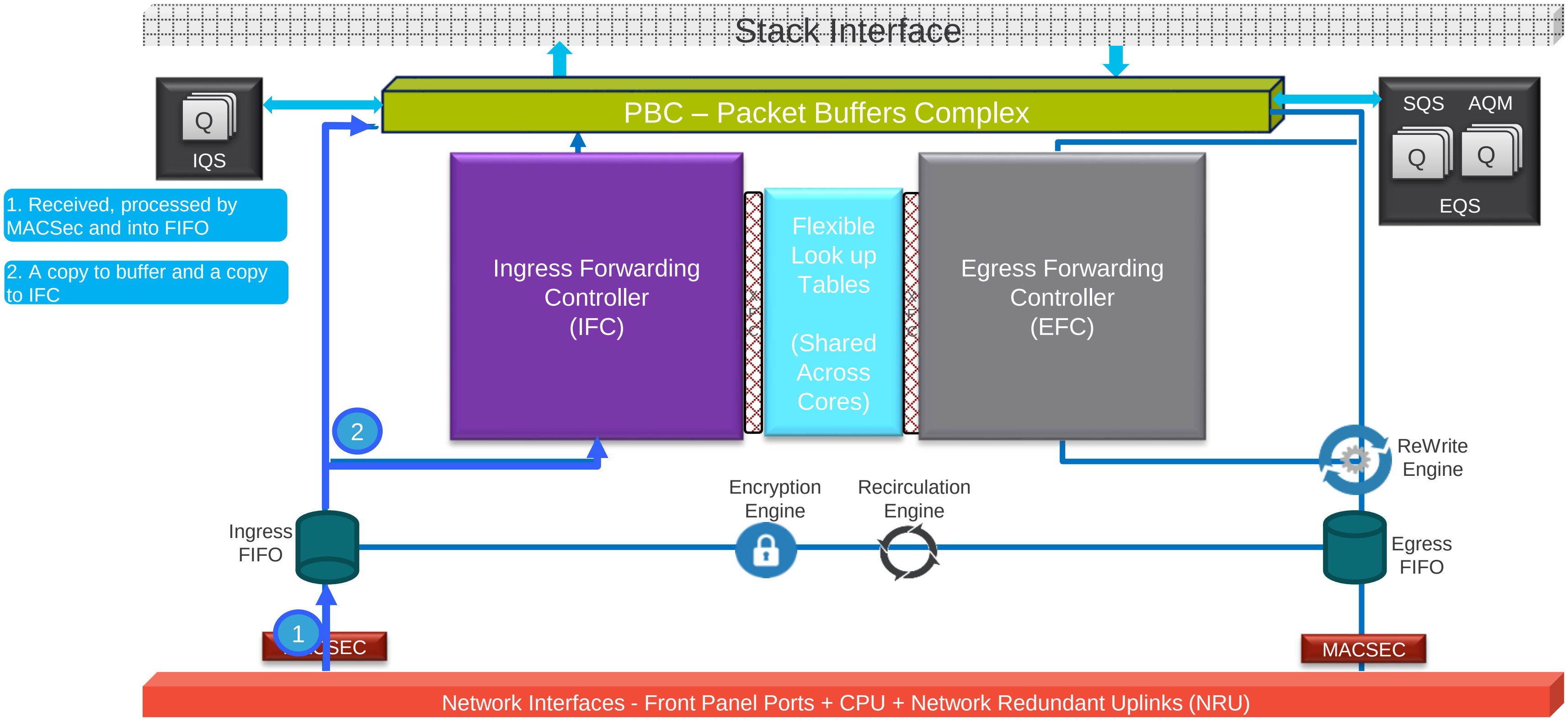
Unicast – within ASIC



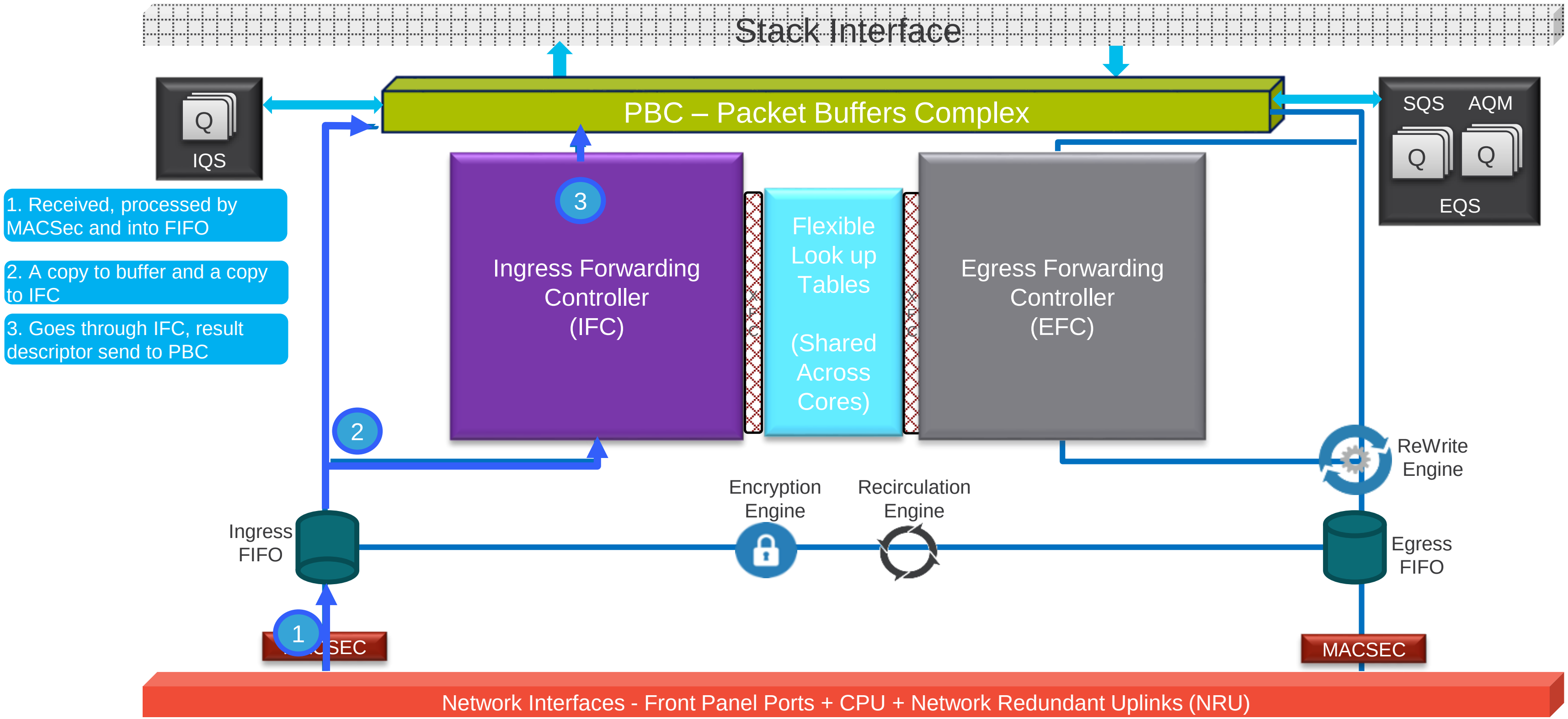
Unicast – within ASIC



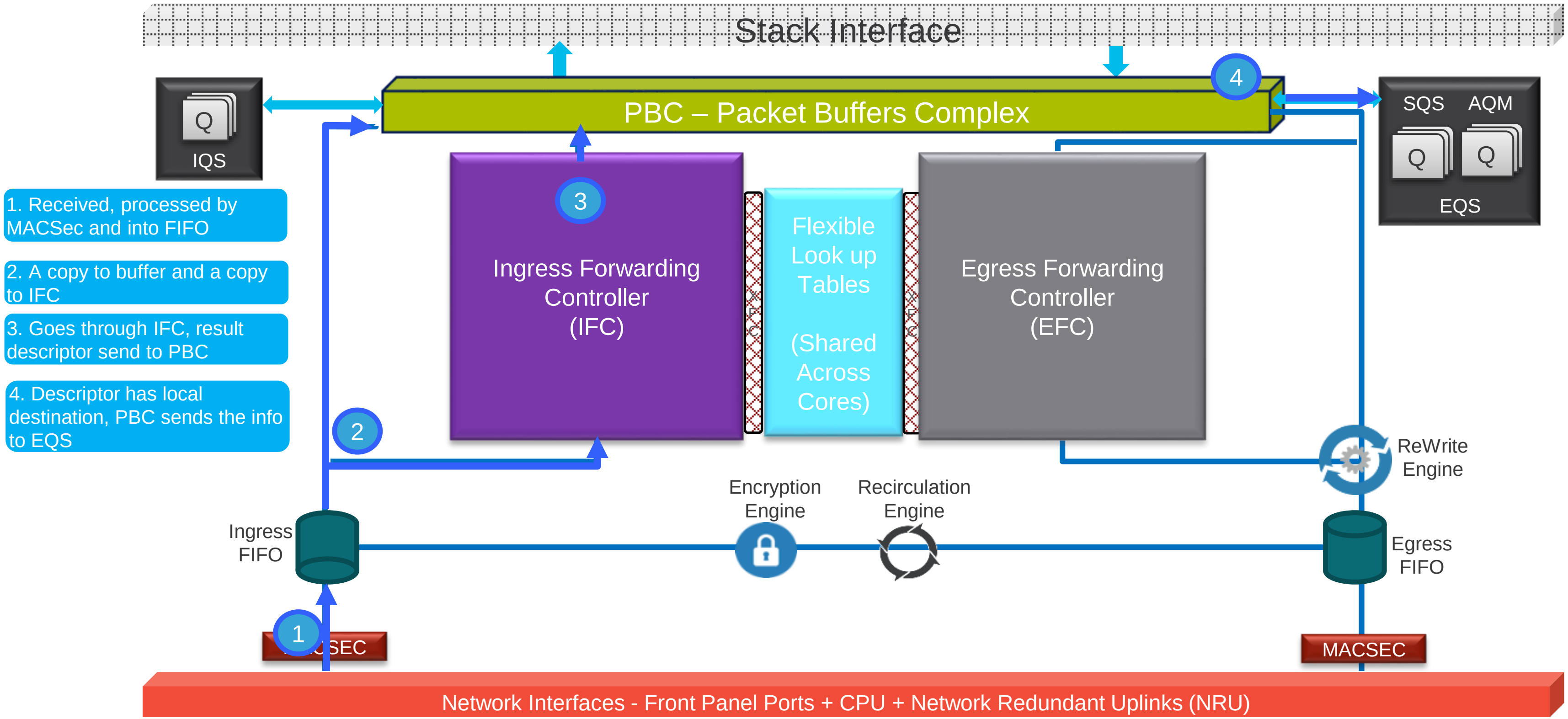
Unicast – within ASIC



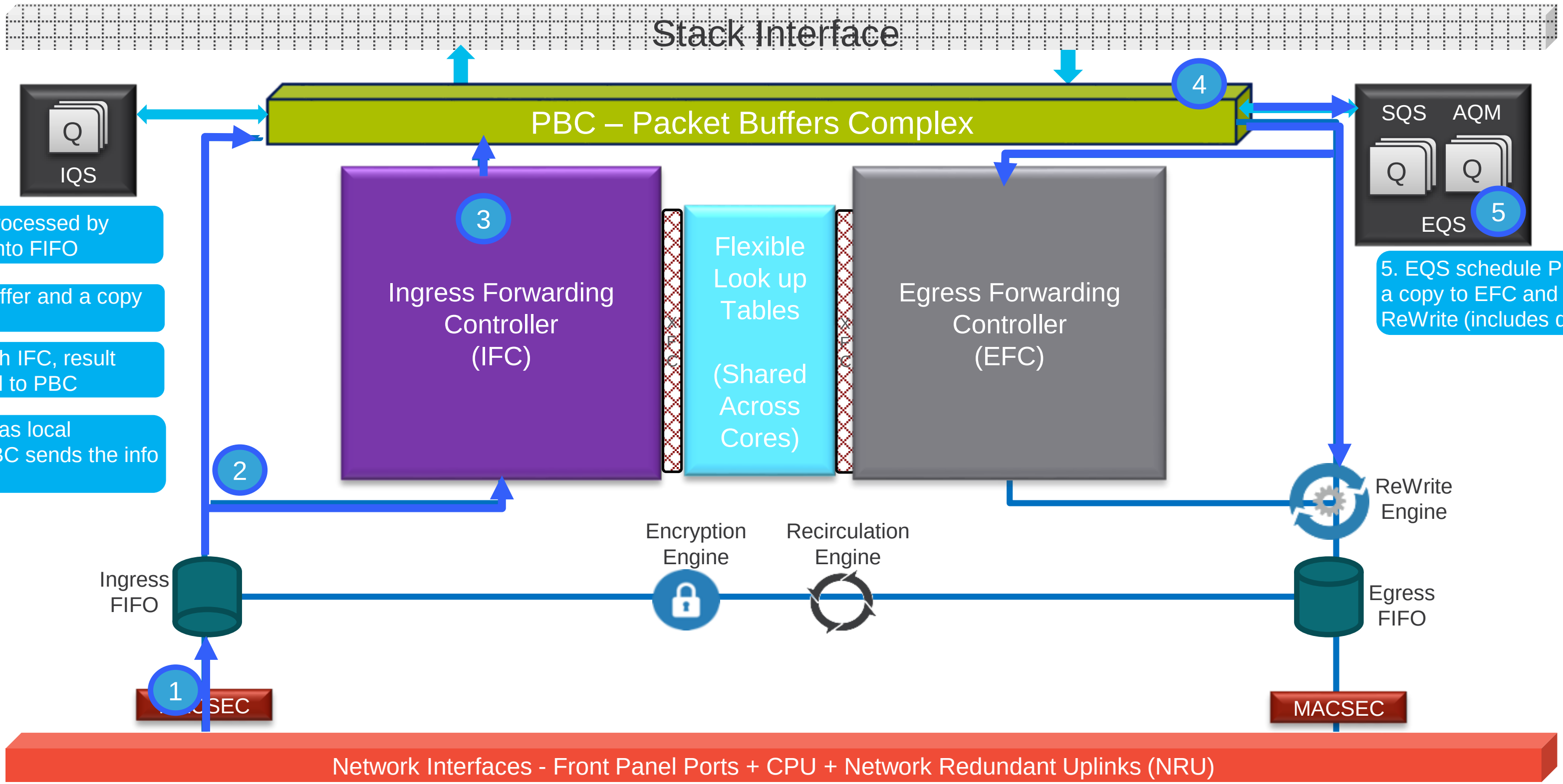
Unicast – within ASIC



Unicast – within ASIC



Unicast – within ASIC



1. Received, processed by MACSec and into FIFO

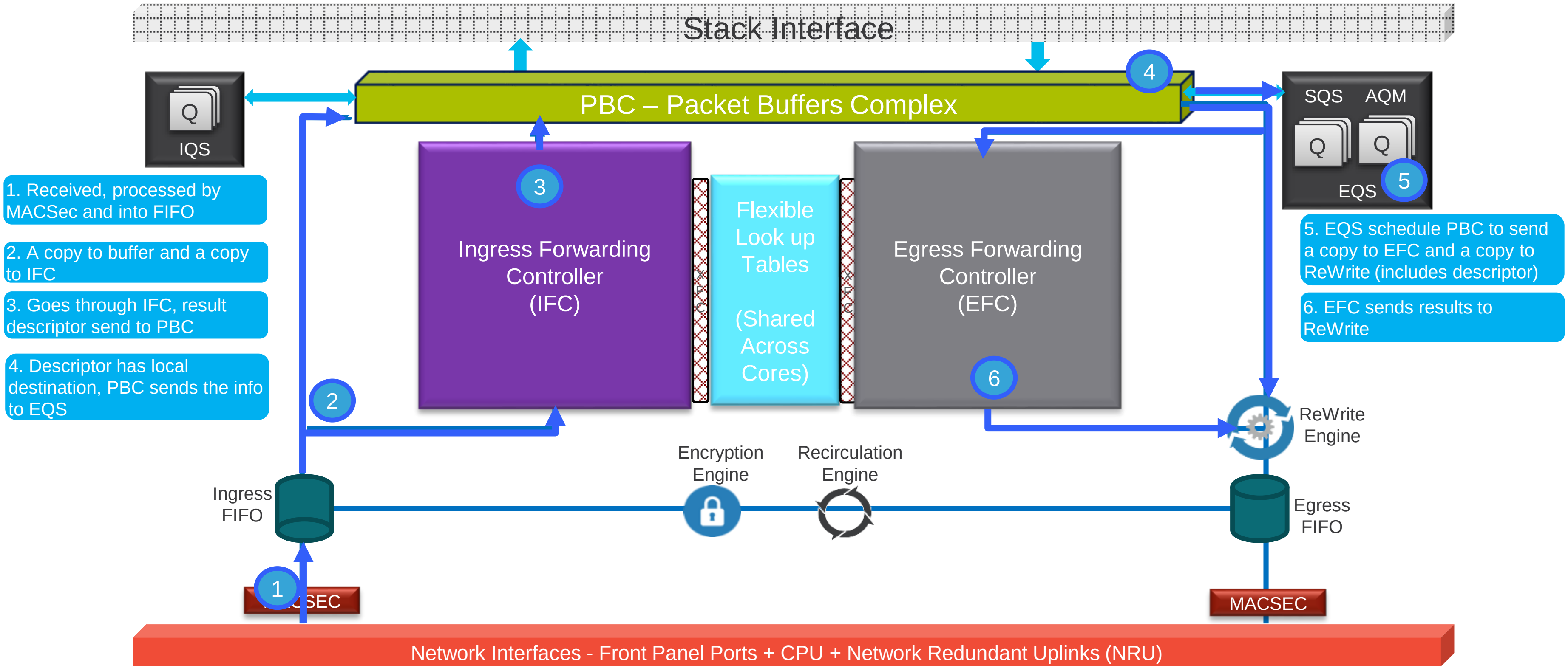
2. A copy to buffer and a copy to IFC

3. Goes through IFC, result descriptor send to PBC

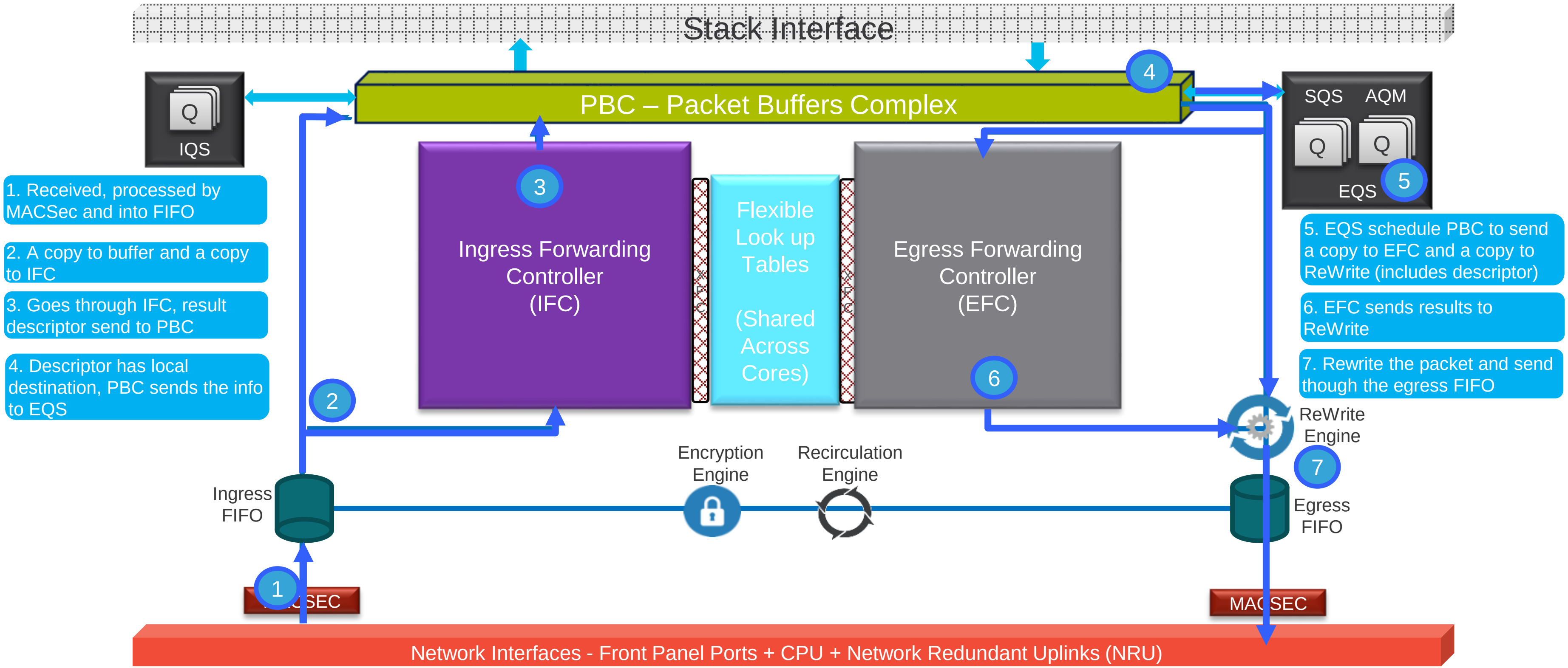
4. Descriptor has local destination, PBC sends the info to EQS

5. EQS schedule PBC to send a copy to EFC and a copy to ReWrite (includes descriptor)

Unicast – within ASIC



Unicast – within ASIC



Multicast

Catalyst 9400 Multicast

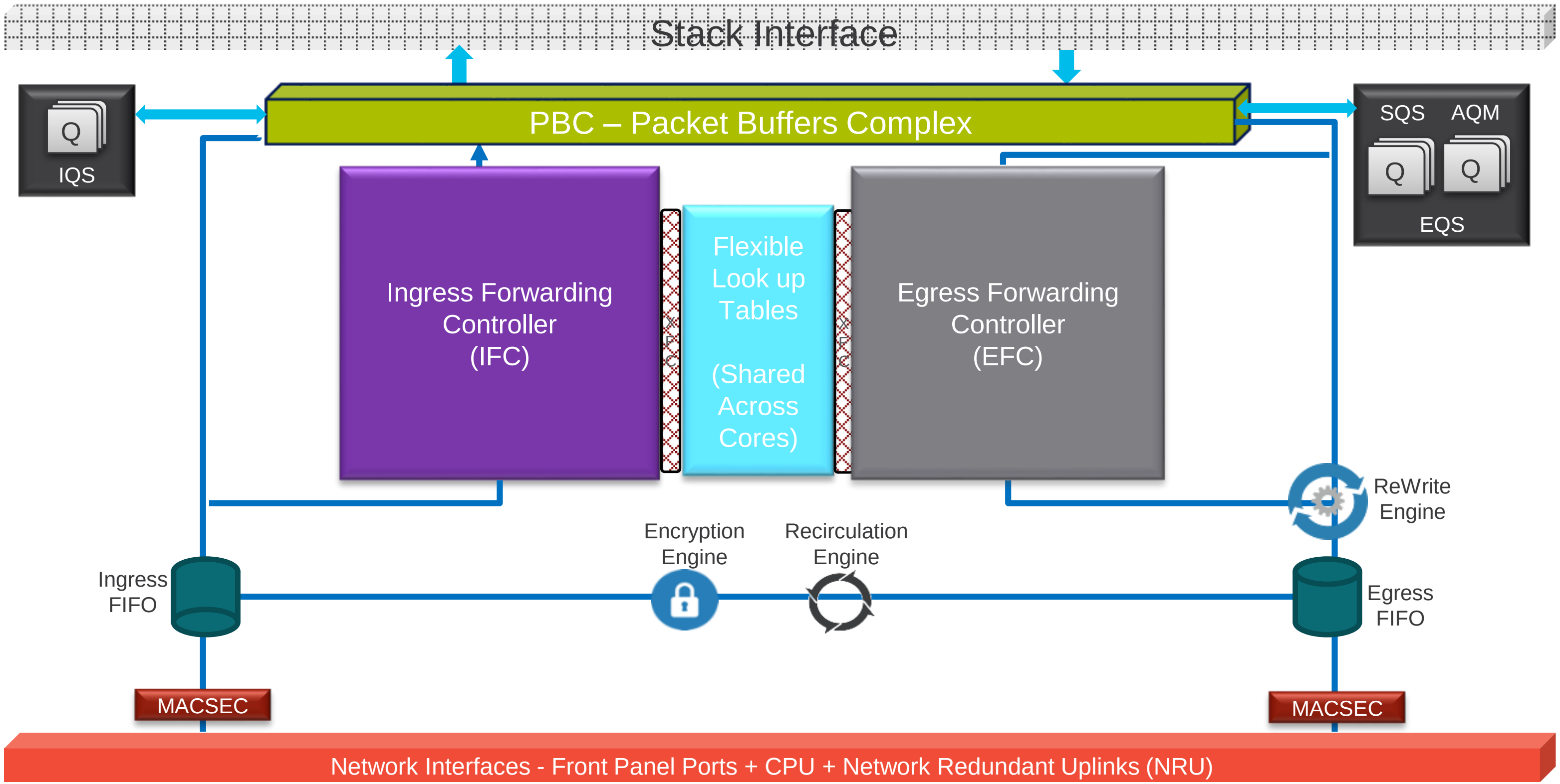
- Performance
  - High L2 and L3 multicast throughput
  - Low latency
- Flexible Scalability
  - Future templates will provide different IGMP snooping group and mroutes
- Optimized Replication
  - Replication are done at the egress
  - Single copy in the buffer memory during replication
- Enhanced Features
  - IP-Based forwarding for IGMP snooping
  - IGMP snooping explicit tracking (For IGMP v3).

FYI

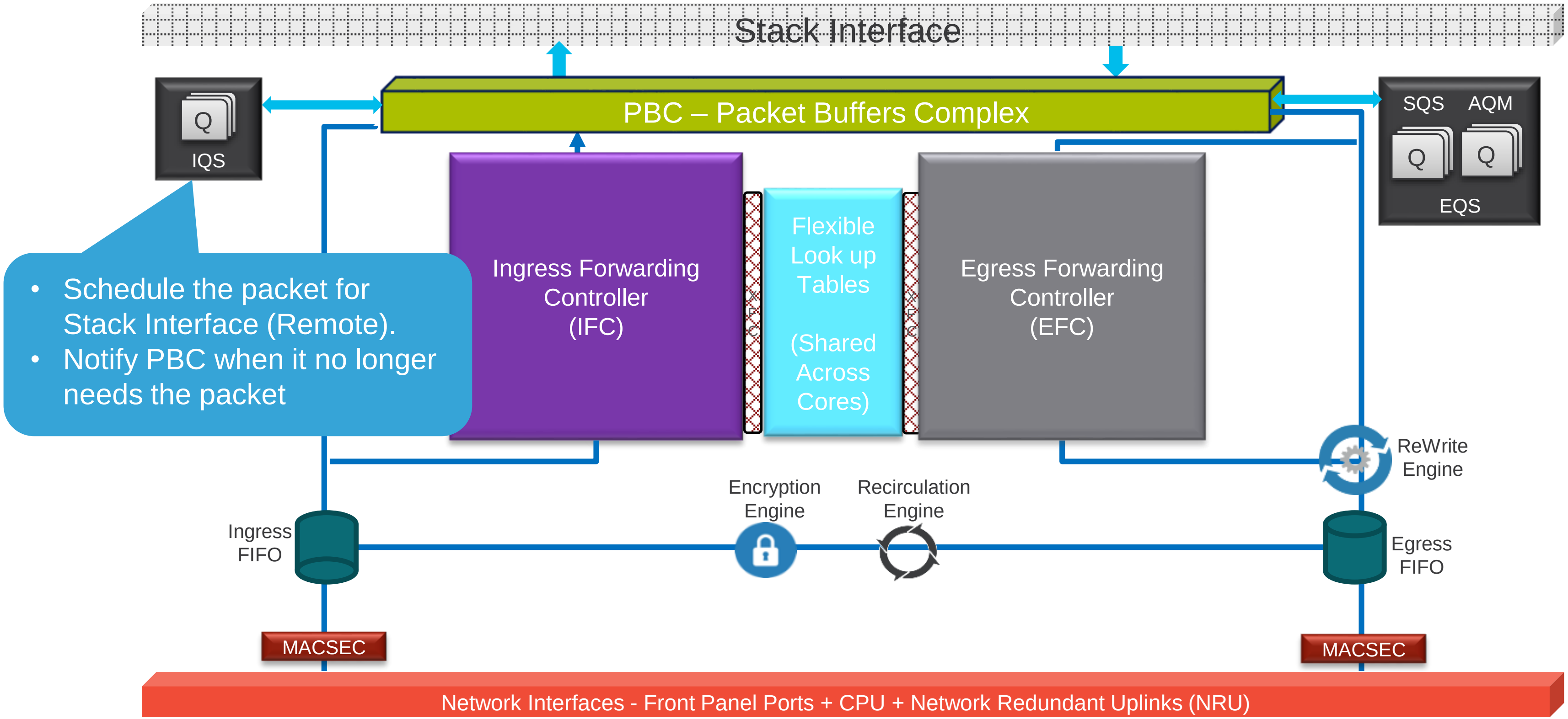
Multicast Features

Features	Catalyst 9K
IGMP (Internet Group Management Protocol)v1, v2, v3	Yes
IGMP snooping (v1, v2, v3)	Yes
MLD (Multicast Listener Discovery) v1, v2	Yes
MLD snooping (v1, v2)	Yes
PIM (Protocol Independent Multicast) SM (Sparse Mode)	Yes
PIM Dense Mode	Yes
PIM SSM (Source Specific Mode)	Yes
PIM Bi-Dir	No (HW capable)

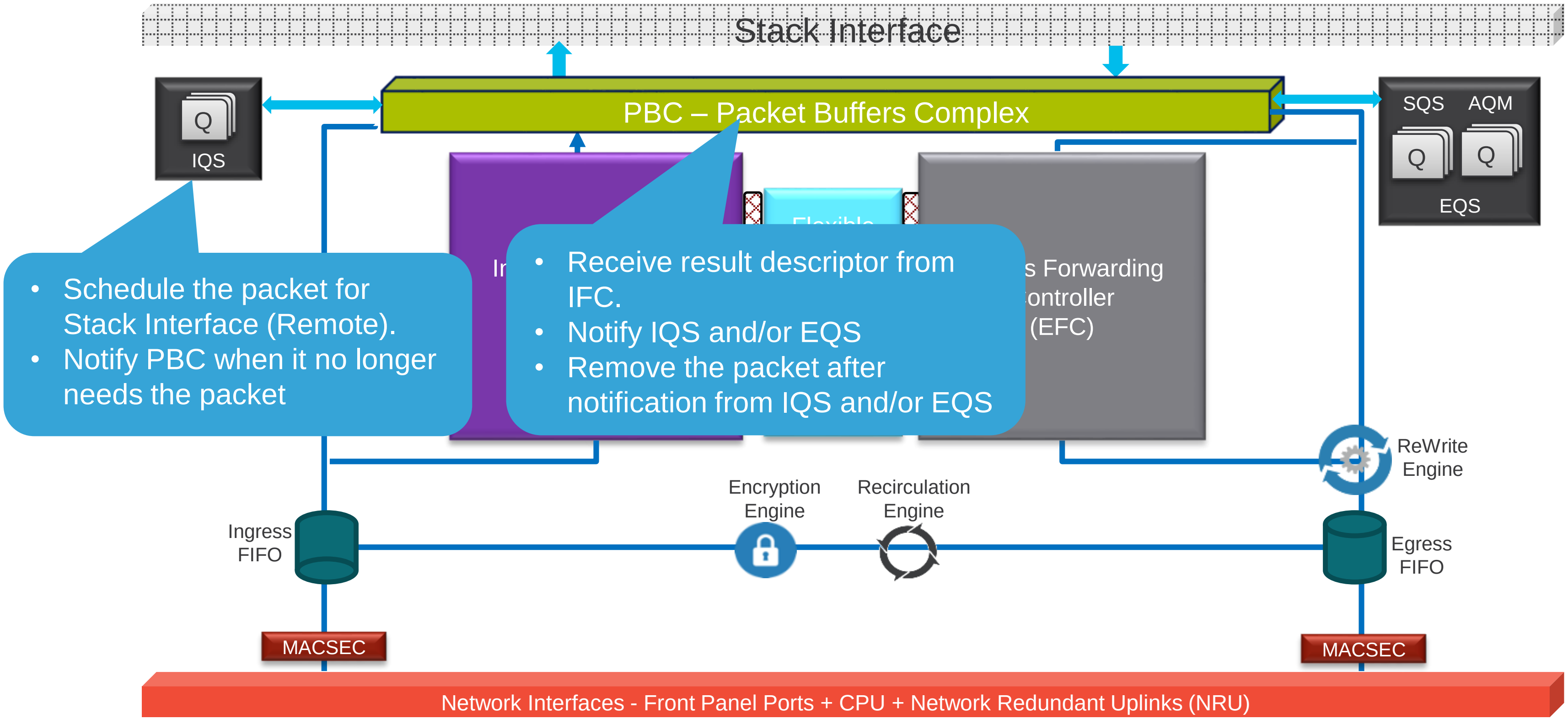
Multicast Key Components



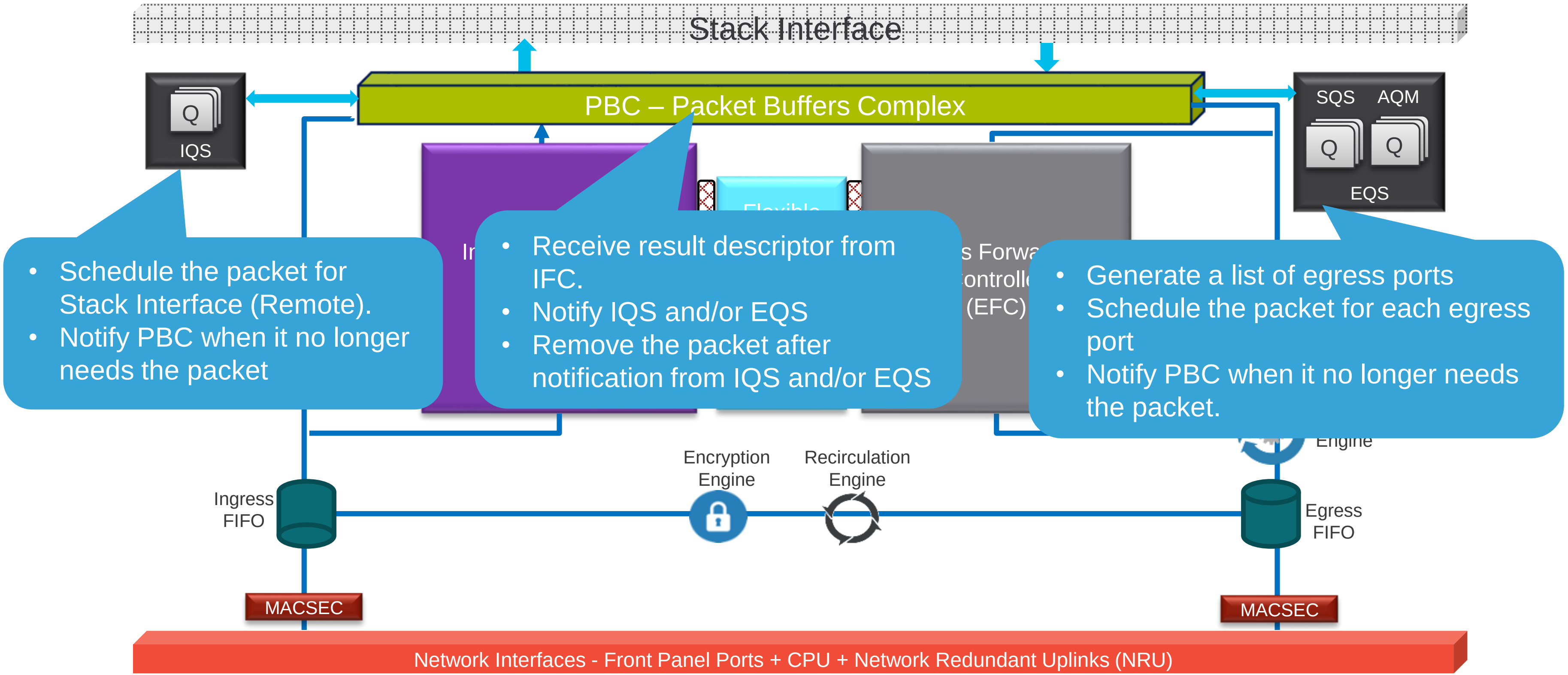
Multicast Key Components



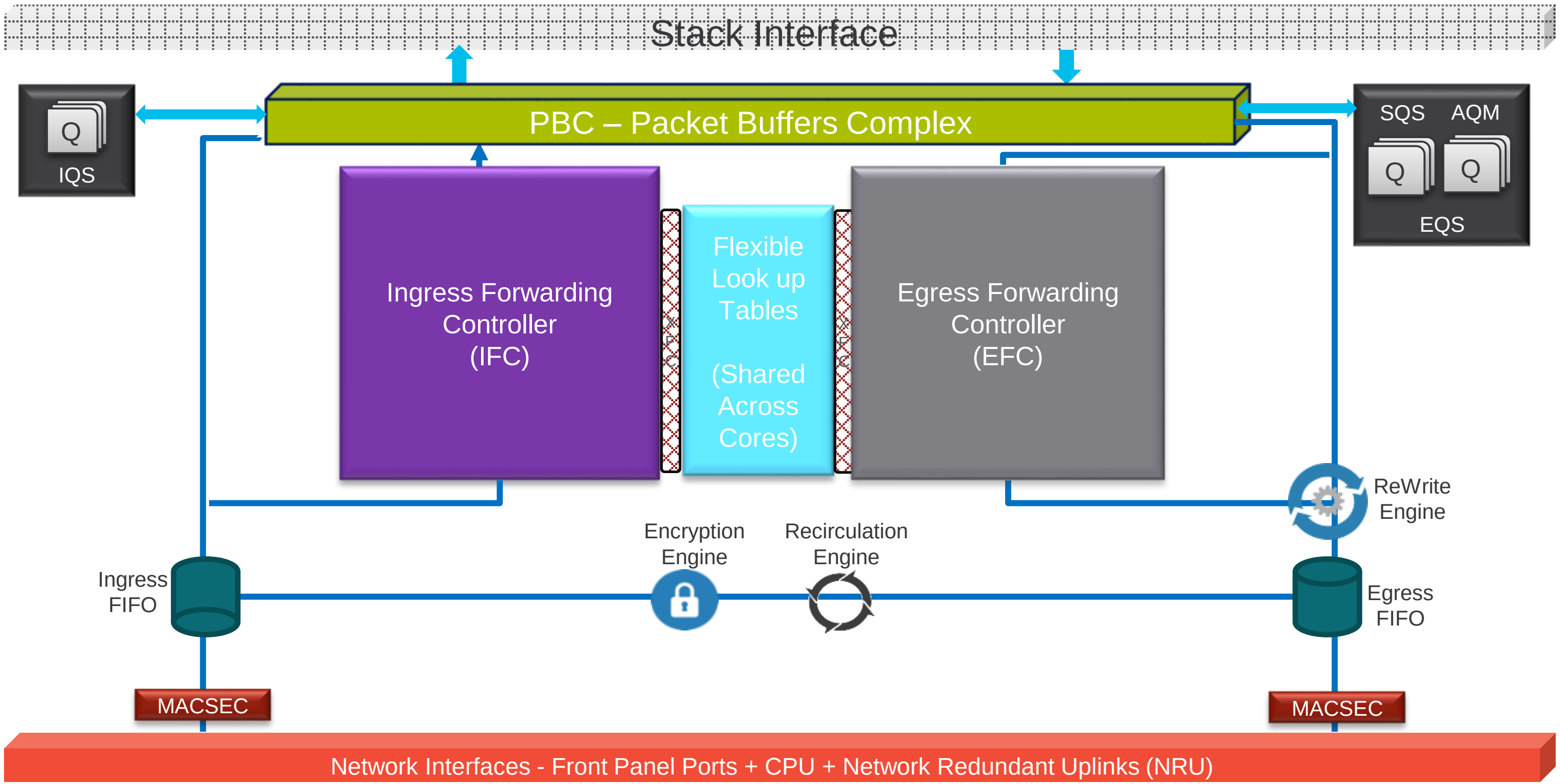
Multicast Key Components



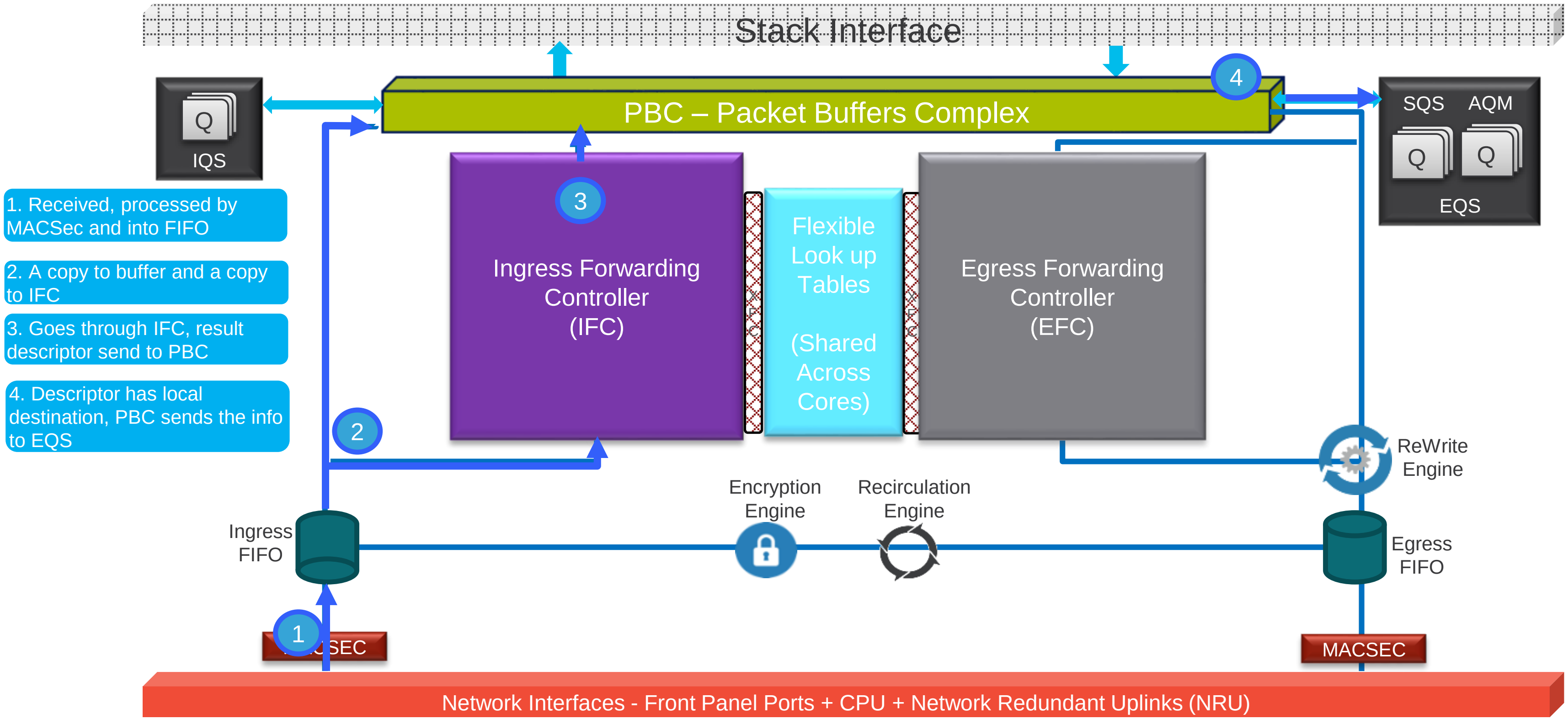
Multicast Key Components



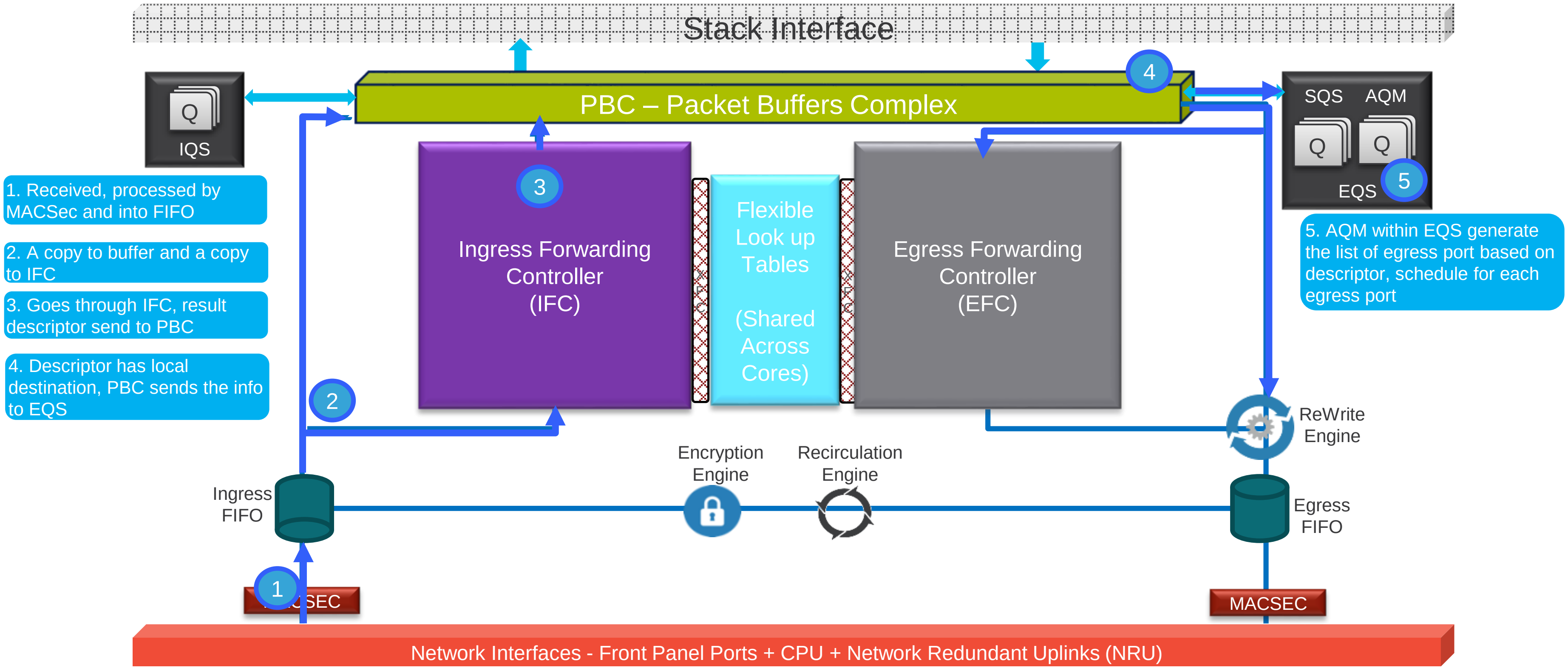
Multicast – Egress Local



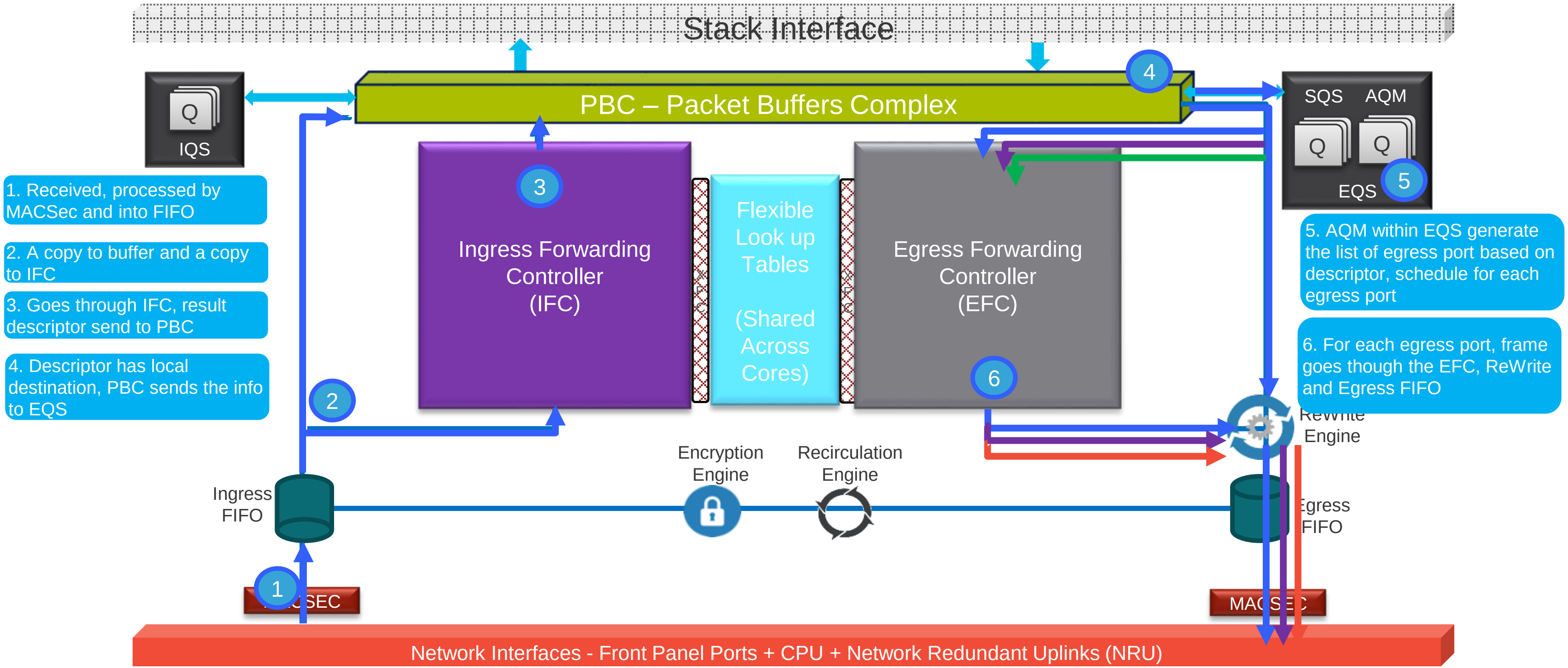
Multicast – Egress Local



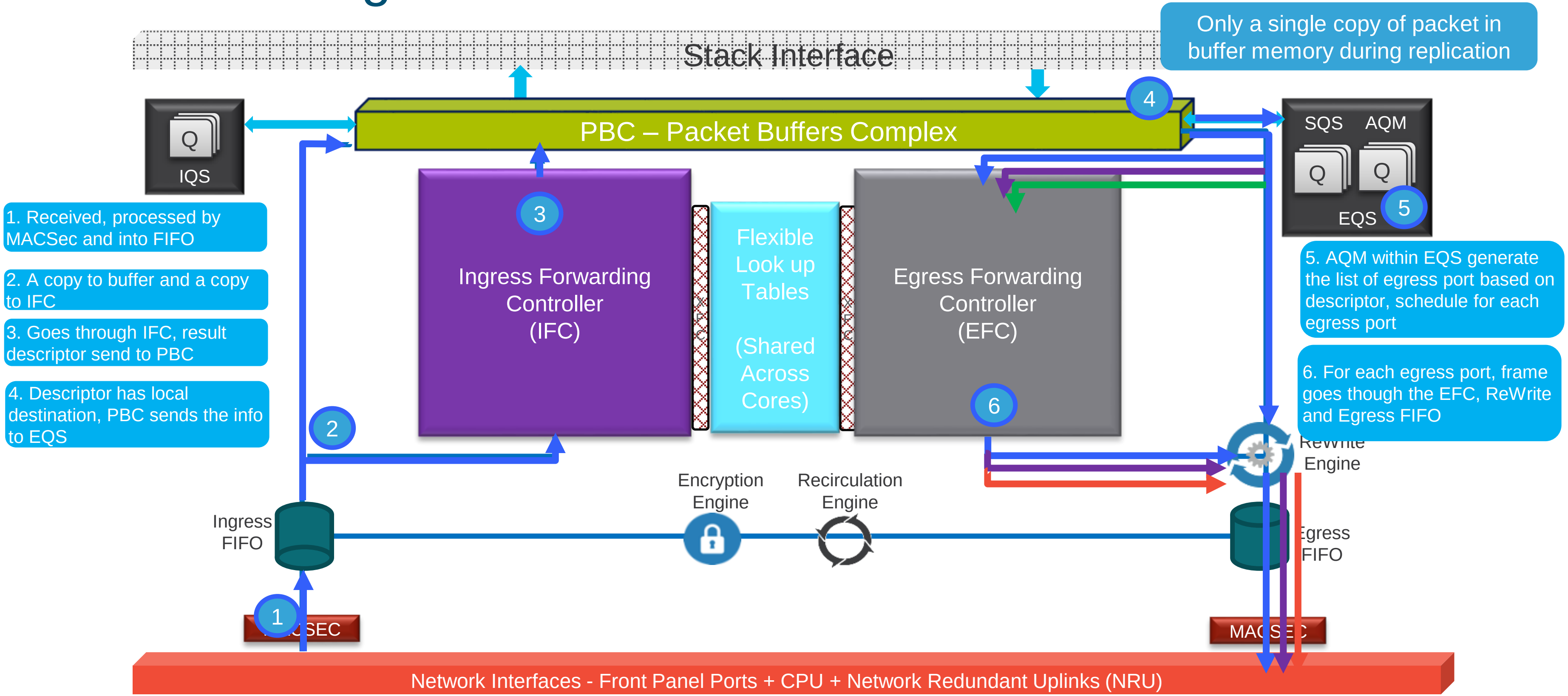
Multicast – Egress Local



Multicast – Egress Local



Multicast – Egress Local



ACL

ACL Types and Features

- Security ACL (MAC, IPv4, and IPv6)
  - PACL: ACL enabled under L2 interface
  - VACL: ACL enabled for L2 VLAN traffic
  - RACL: ACL enabled for routed traffic
  - GACL: ACL for Clients group
  - SGACL: ACL for CTS/SGT
  - WCCP-Egress
- ACL for QoS classification and Policing (including CoPP)
- Service ACLPolicy
  - PBR/NAT/WCCP-Ingress
  - Netflow ACL
  - SPAN
  - MACSec
  - User ACL to redirect traffic (CoPP)
  - Tunnel
  - LISP

Security ACLs

```
interface gigabitethernet1/1
  ip access-group PACL-1 in
  ip access-group PACL-2 out
  switchport access vlan 100
vlan access-map VACL-map
  match ip address VACL-1
  action forward
vlan filter VACL-map vlan-list 100
interface Vlan100
  no shutdown
  ip access-group RACL-1 in
  ip access-group RACL-2 out
  ip address 100.1.1.1/24
```

PACL: Direction indicated in CLI

VACL: VLAN ACL for Both Ingress and Egress; Apply to Pre- and Postrouted Traffic

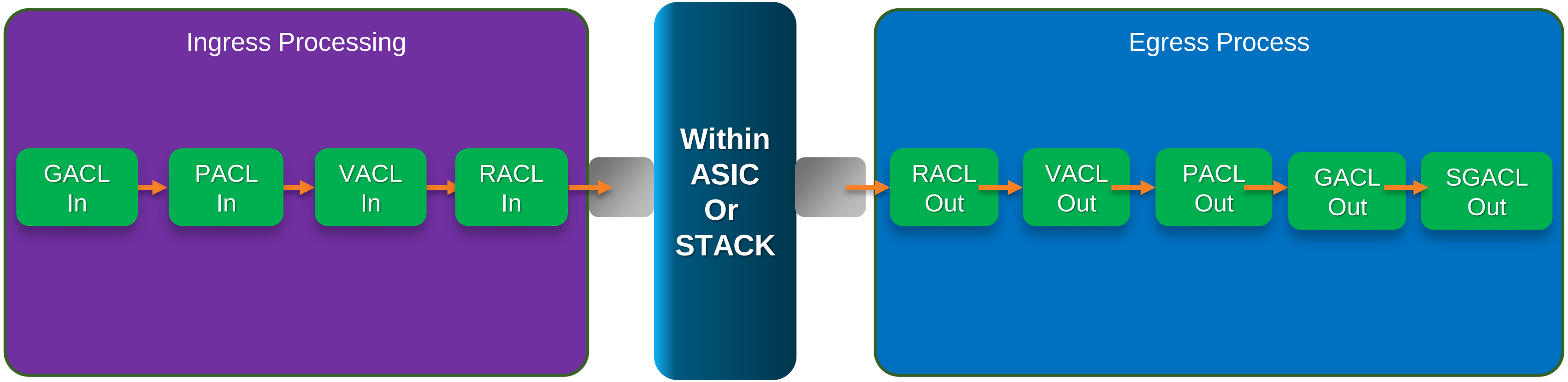
RACL: Routed ACL; Apply Only to Traffic That Requires Routing; Direction Indicated in the CLI

ACL Type	Attach Point	Direction
PACL	L2 interface	IN OUT
VACL	VLAN	Always IN & OUT
RACL	L3 interface, L3 PortChannel, sub-interface SVI	IN OUT

FYI

Security ACL Processing Order and Priority

- The following is a conceptual illustration. In the ASIC the lookup for different types of ACL takes place concurrently.
- A packet is dropped if it hits the deny rule in any of these types of ACLs.
- RACL is applied only to traffic that is L3 forwarded.



Sharing ACE

- Each ACL policy is reference by a label.
- Same ACL policy (Security ACL like GACL, PACL, VACL, RACL and QoS ACL) is applied to multiple interfaces or VLAN, it uses the same label.
- Label sharing is within the core
- Ingress and egress use different label

```
ip access-list extended ip-list-1
 deny   ip 100.1.1.0 0.0.0.255 200.1.1.0 0.0.0.255
 deny   ip 100.1.2.0 0.0.0.255 200.1.2.0 0.0.0.255
 deny   ip 100.1.3.0 0.0.0.255 200.1.3.0 0.0.0.255
 permit ip any any

interface GigabitEthernet1/0/1
 ip access-group ip-lists-1 in
interface GigabitEthernet1/0/2
 ip access-group ip-lists-1 in
interface GigabitEthernet1/0/3
  ip access-group ip-lists-1 in
```

Label	
101	Gi1/0/1
101	Gi1/0/2
101	Gi1/0/3
101	IPv4 ACL ip-list-1 in

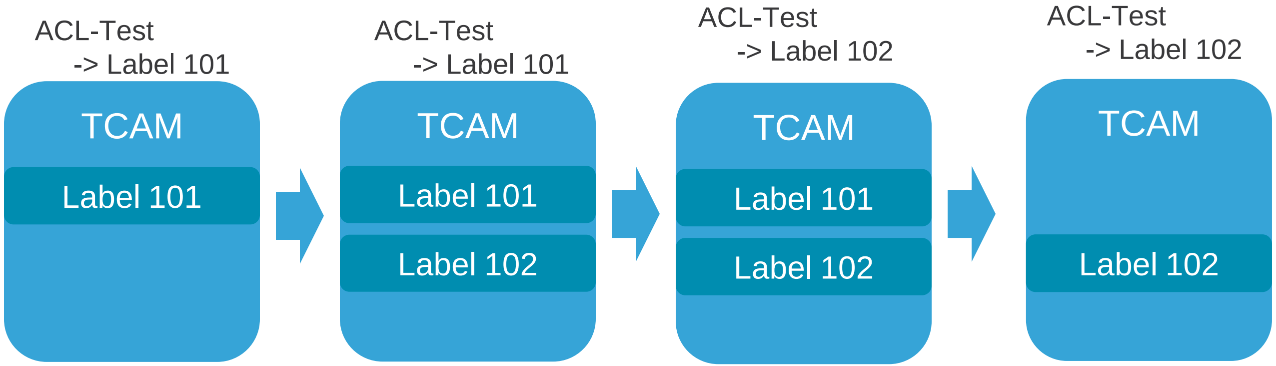
FYI

Resource Utilization

```
C9400#show platform hardware fed active fwd-asic resource tcam utilization
CAM Utilization for ASIC Instance [0] Table         Max Values        Used Values
--------------------------------------------------------------------------------
 Unicast MAC addresses                               65536/512          17/21
 IGMP and Multicast groups                           8192/512           0/0
 L2 Multicast groups                                 8192/512           0/0
 Directly or indirectly connected routes             16384/65024          0/14
 NAT/PAT SA address and Port                            0                 0
 QoS Access Control Entries                          18432                 0
 Security Access Control Entries                     18432               114
 Ingress Netflow ACEs                                1024                 8
 Policy Based Routing ACEs                           2048                 0
 Egress Netflow ACEs                                 2048                 0
 Input Microflow policer ACEs                         512                 0
 Output Microflow policer ACEs                       1024                 7
 Flow SPAN ACEs                                      1024                 0
 Control Plane Entries                               1024               200
 Tunnels                                             1024                17
 Lisp Instance Mapping Entries                       1024                 3
 Input Security Associations                          512                 0
 Output Security Associations and Policies            512                 5
 SGT_DGT                                             8192/512           0/0
 CLIENT_LE                                           4096/256           0/0
 INPUT_GROUP_LE                                      1024                 0
 OUTPUT_GROUP_LE                                     1024                 0
 Macsec SPD                                           256                 2
```

Hitless ACL update

```
Modified ACL-Test

SW(config)#ip access-list extended ACL-Test
SW(config-ext-nacl)#105 permit ip host 10.1.1.1 host 20.1.1.1
SW(config-ext-nacl)#
```

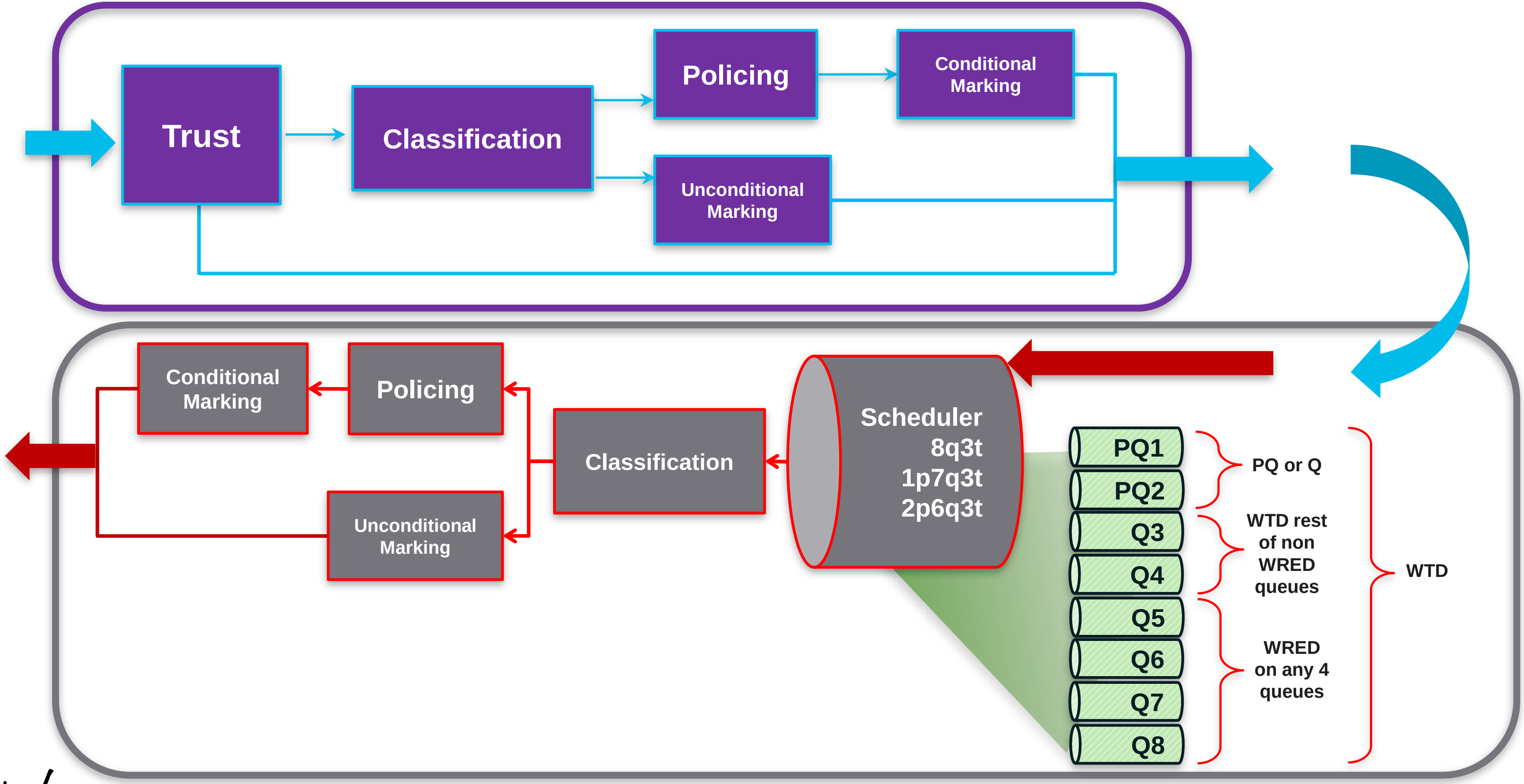


Requirement:
TCAM has enough space

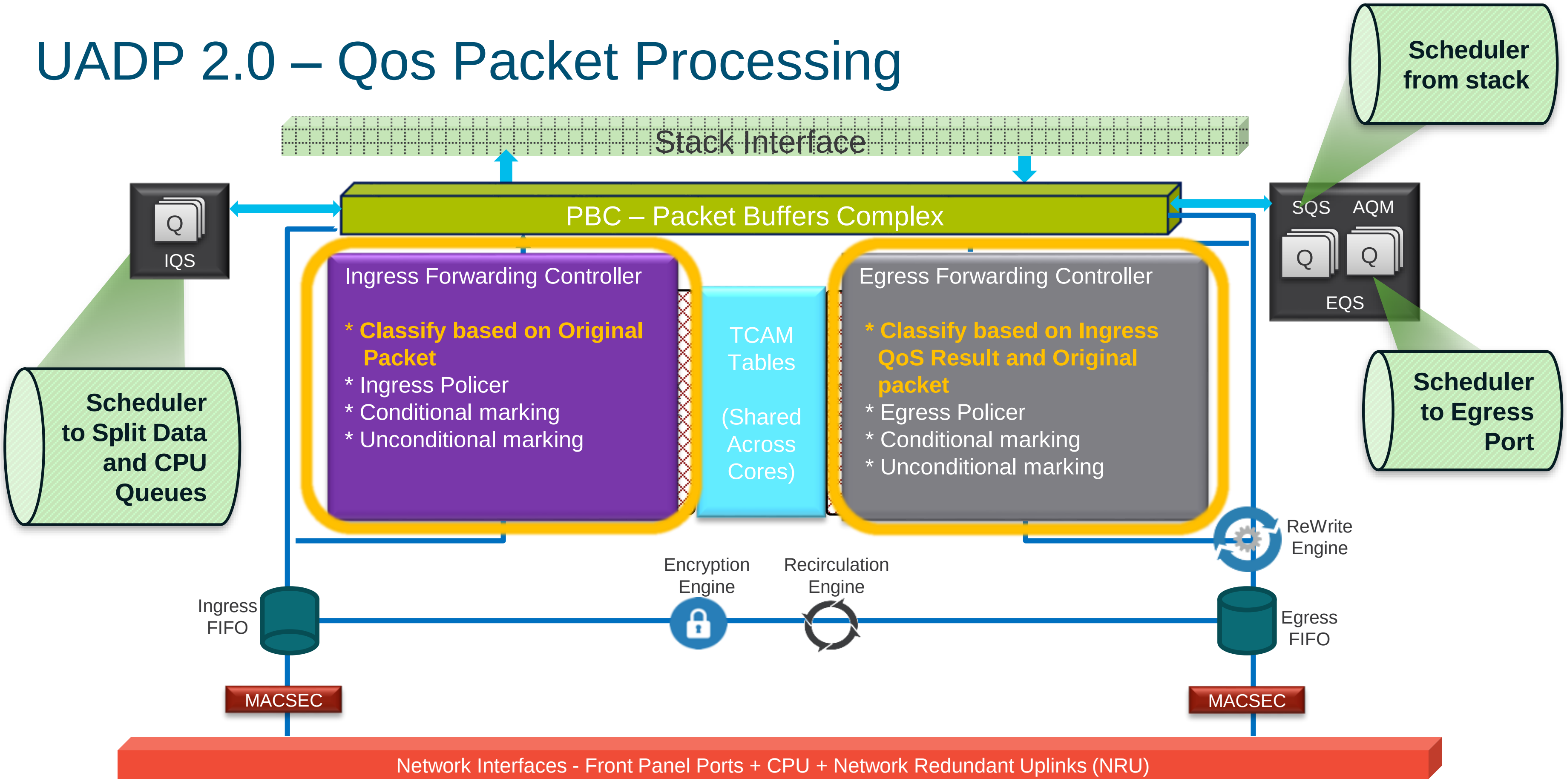
Result:
No outage during ACL update

QoS

QoS Fundamental

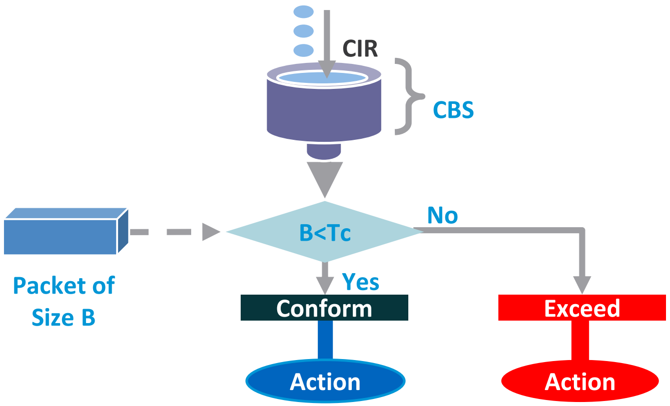


UADP 2.0 – Qos Packet Processing



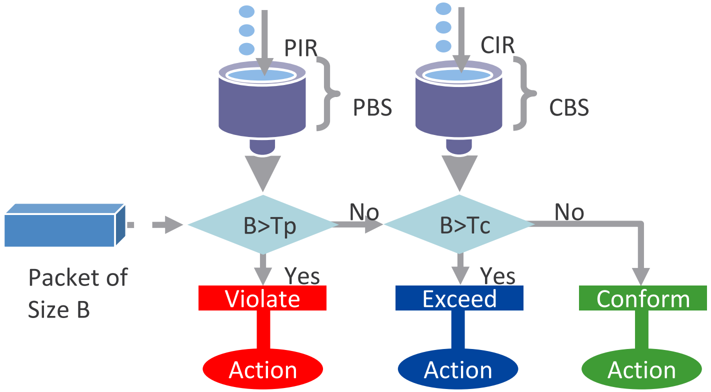
Policing

1 Rate 2 Color




```
police cir 100000000 bc 3125000 conform-action set-dscp-transmit af41 exceed-action drop
```

2 Rate 3 Color




```
police cir percent 10 pir percent 50 conform-action transmit exceed-action set-dscp-transmit af11 violate-action drop
```

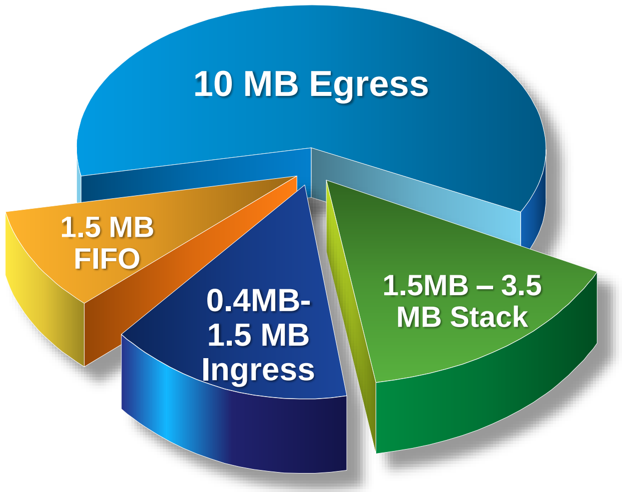

Packet Buffer

Types:

- Ingress Buffers: For ingress queues and scheduler (IQS) (dedicated and shared)
- Egress Stack Buffers: For Egress Stack Queues and Scheduler (SQS) (dedicated and shared)
- Egress Port Buffers: For Egress port queues (AQM)
- Temporary Buffers: For packet from the Stack or CPU.

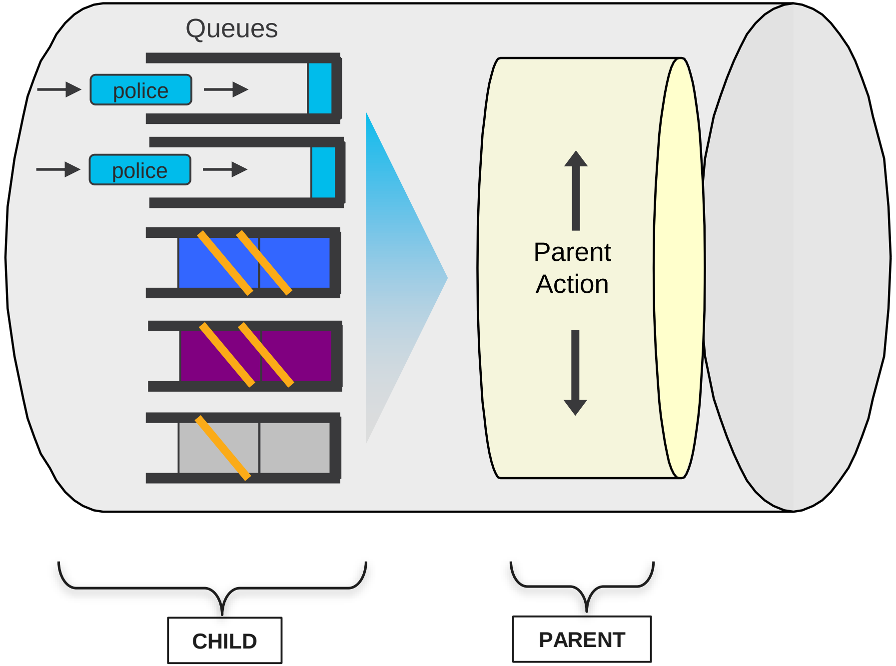
Allocation:

- Use dedicated first then shared.



16MB per Core
32MB per ASIC

2-Level HQoS



Child	Parent
BW, Policing	Shape
Marking	Policing
Policing	Marking

High Availability

SSO – Stateful Switch Over

AP | PC/Laptop | IP Phone

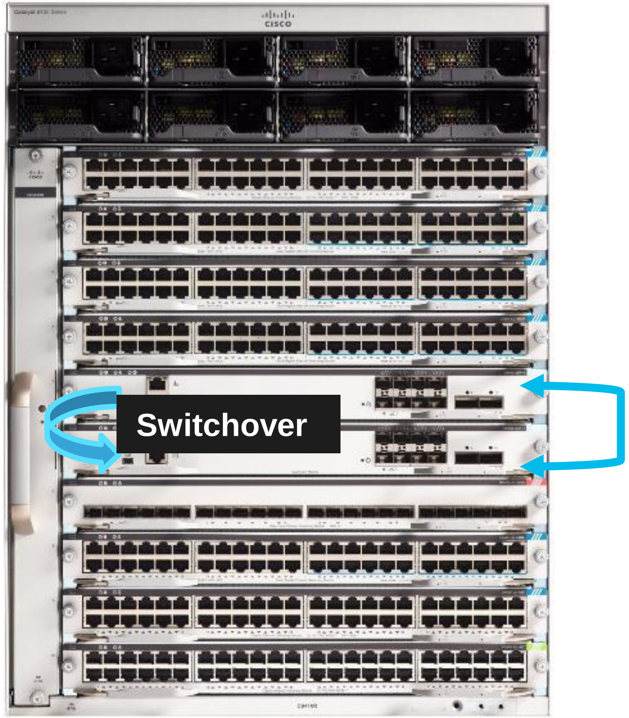
Links Stay up
Continue Communication

LED Lights

Lights Stay on

Sup Uplinks stay up

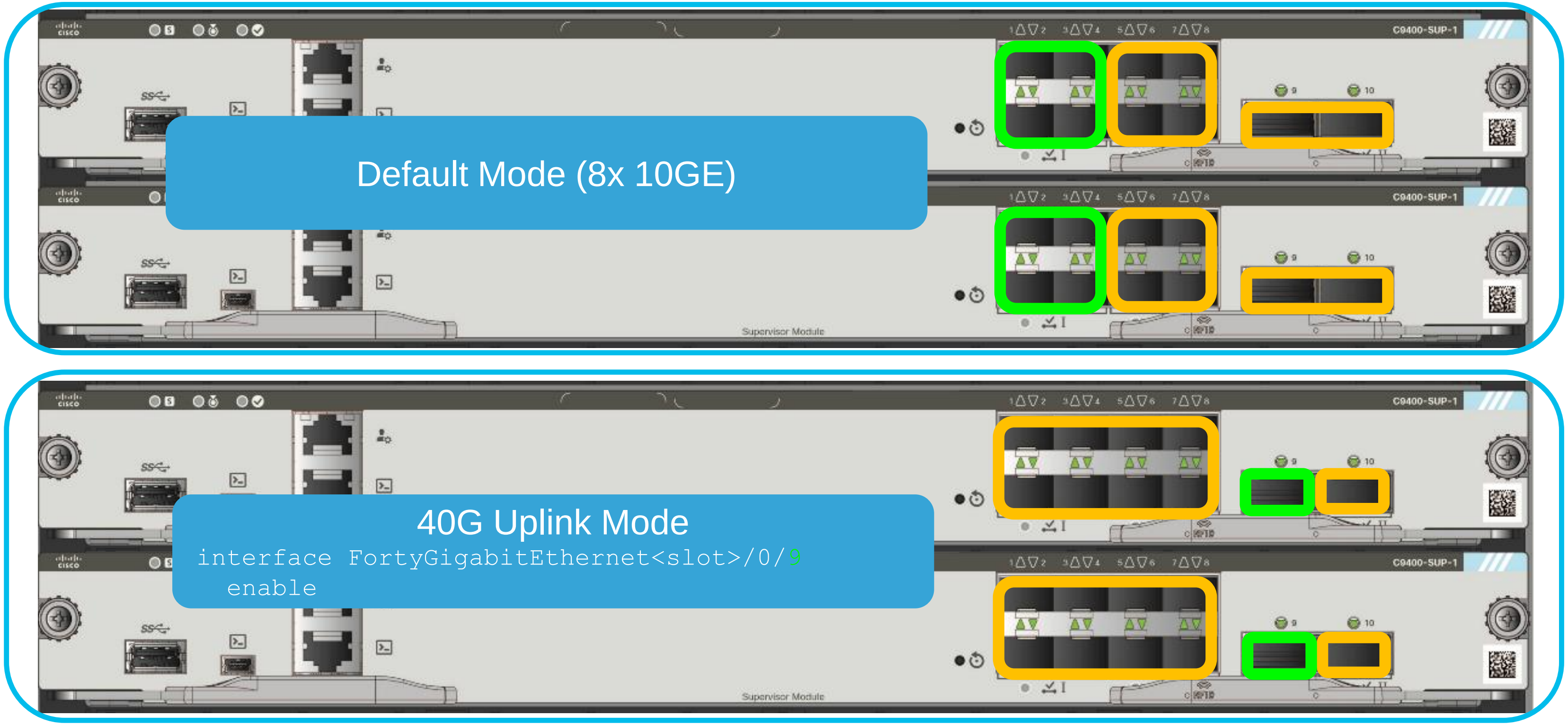
Less than 200ms traffic interruption



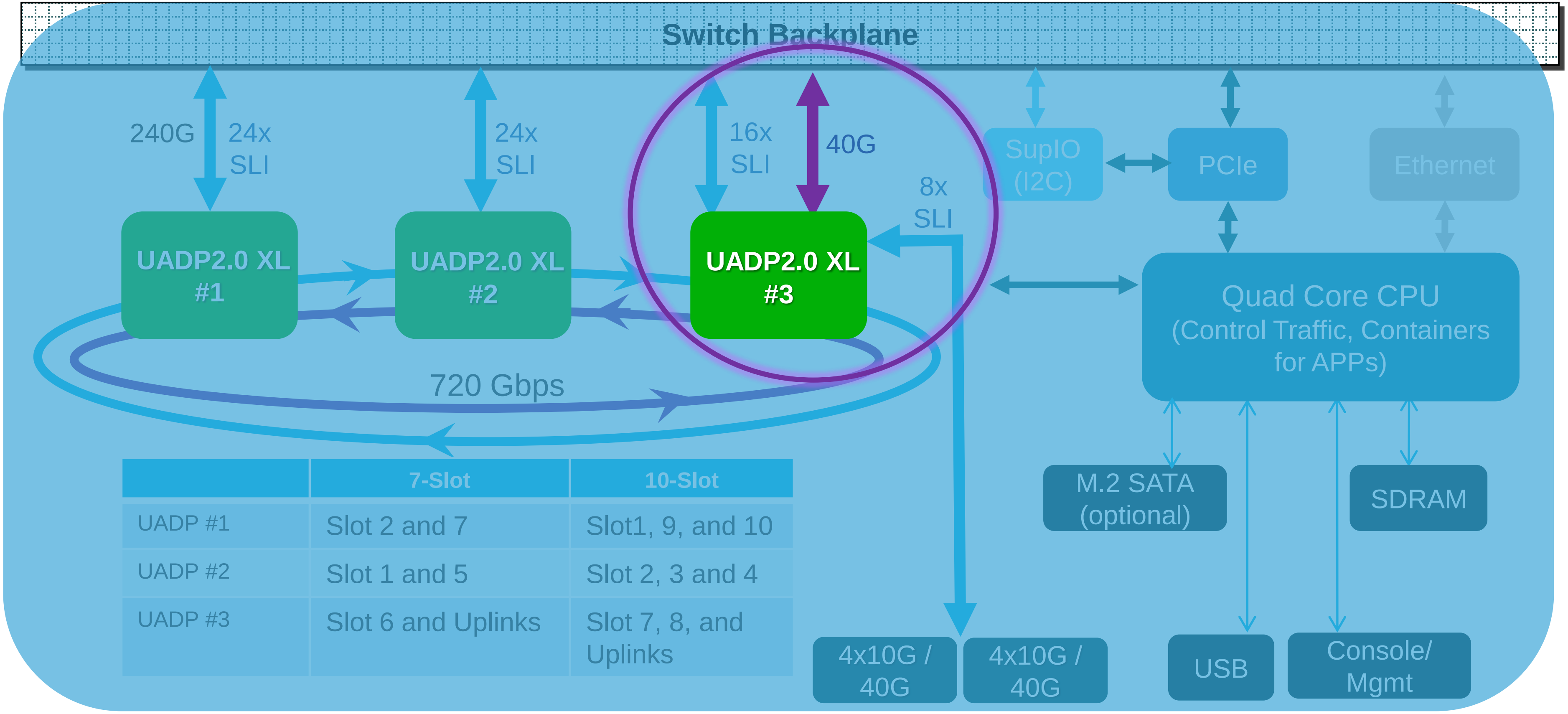
Default Redundancy Mode

Continue Synchronization from Active to Standby

Sup-1 Uplink Redundancy – Dual Sups

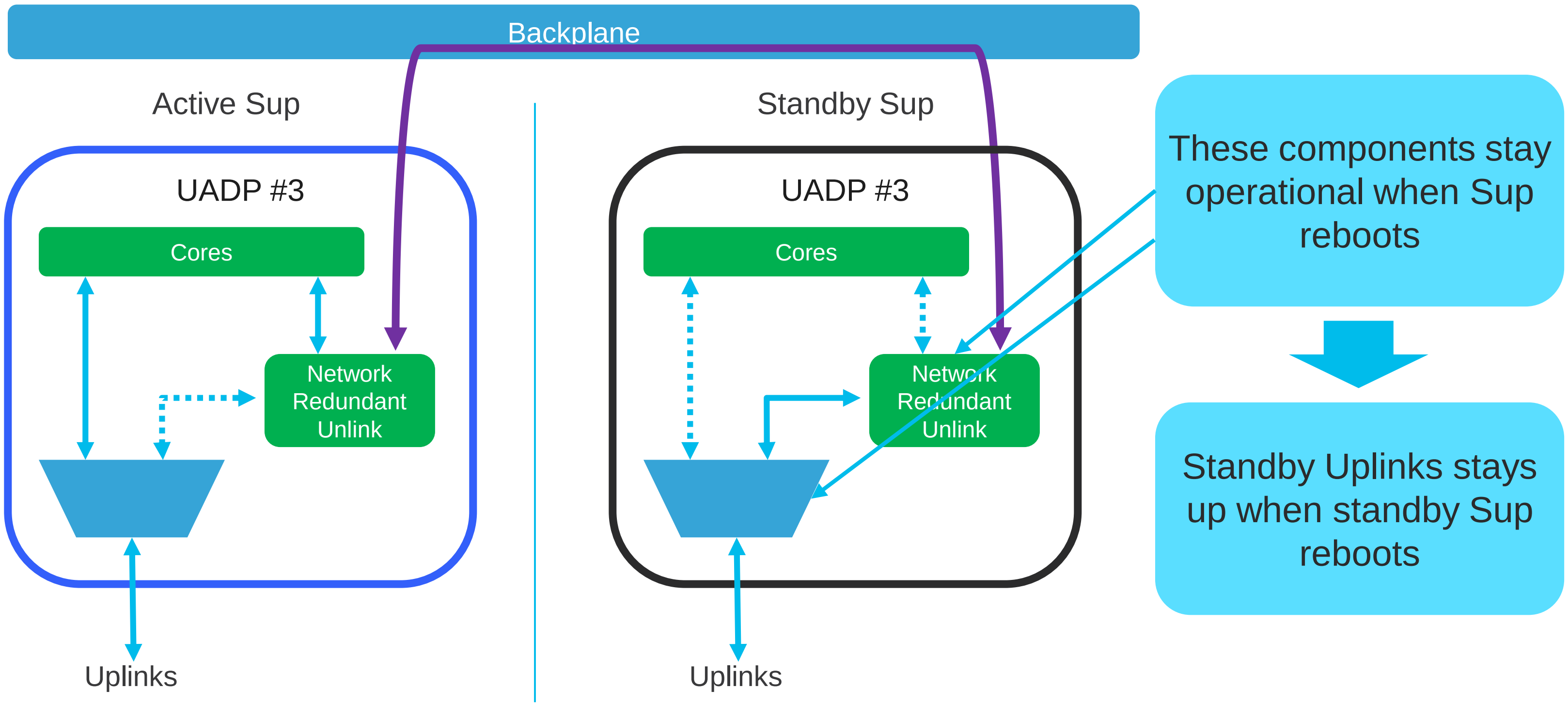


Redundant Uplinks



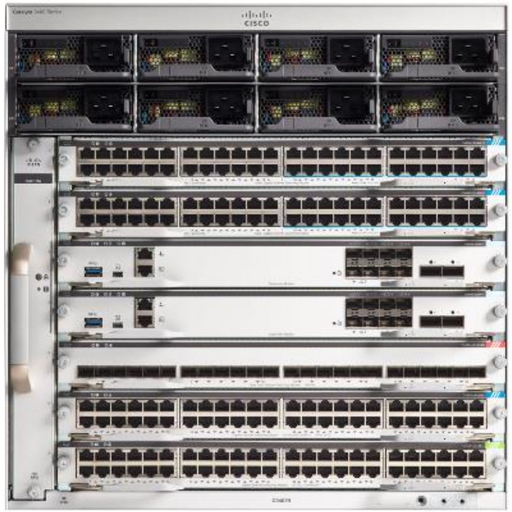
	7-Slot	10-Slot
UADP #1	Slot 2 and 7	Slot1, 9, and 10
UADP #2	Slot 1 and 5	Slot 2, 3 and 4
UADP #3	Slot 6 and Uplinks	Slot 7, 8, and Uplinks

Redundant Uplinks



Conclusion

Catalyst 9400

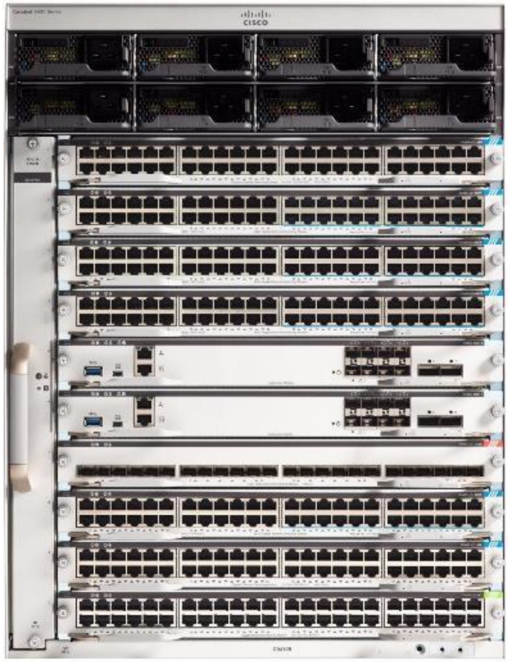


Open IOS-XE

480G Slot BW

High Availability/ Redundancy

100W PoE Ready



Supervisor	Access Linecards	Core Linecards	Power Supply
Sup-1: 80G/Slot Access Optimized	24xmGig + 24xUPOE* 48xUPoE 48xData	24x 10G SFP+* 24x1G SFP*	3200W AC

*not available at FCS

Catalyst 9K enables the New Era of Networking

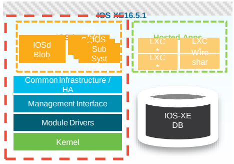

X86 Based CPU


Larger Storage

Usability & Operational Efficiency

Catalyst 9K Family

UADP 2.0



IOS XE 16.6

Foundational Components

Encrypted Traffic Analytics (ETA)	Resiliency with Patching & GIR	Programmability & Automation
Standards Leadership .3bt, .3bz, etc	Real time Streaming Telemetry	SD Access
Trustworthy Systems	3rd Party App Hosting	Fabric Enabled Wireless

Catalyst 9K – Built to see you through Next Decade

Continue Your Education

- Cisco Live Sessions:
  - BRKARC-3438: Cisco Catalyst 3850 and 3650 Series Switching Architecture (Tue 8 am; Wed 8 am)
  - BRKARC-3445: Cisco Catalyst 4500E Switch Architecture (Thur 8:30 am)
  - BRKARC-3465: Cisco Catalyst 6800 Switch Architecture (Tue 4 pm; Thur 8:30 am)
  - BRKARC-3863: Catalyst 9300 Switching Architecture (Thur 10:30 am)
  - BRKCRS-2410: Cisco Network Data Platform for Campus Networks (Tue 1:30pm; Wed 8am)
- Labs
  - LTRCRS-2007: Catalyst Access Layer Innovations: Hands On Lab (Tue 1 p.m.)
- Lunch & Learn
  - TTGEN-1002 (Tue 12 pm)

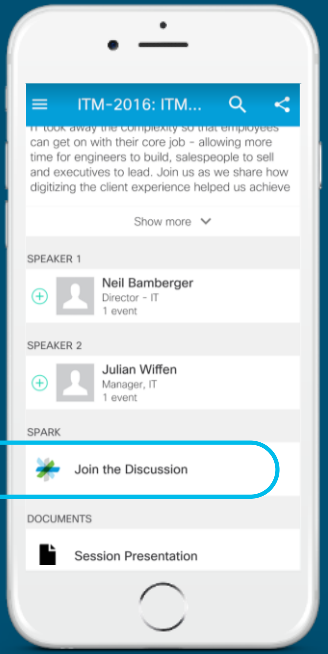
Cisco Spark

Questions?

Use Cisco Spark to communicate with the speaker after the session

How

1. Find this session in the Cisco Live Mobile App
2. Click “Join the Discussion”
3. Install Spark or go directly to the space
4. Enter messages/questions in the space



cs.co/ciscolivebot#BRKARC-3873

Complete Your Online Session Evaluation

- Please complete your Online Session Evaluations after each session
- Complete 4 Session Evaluations & the Overall Conference Evaluation (available from Thursday) to receive your Cisco Live T-shirt
- All surveys can be completed via the Cisco Live Mobile App or the Communication Stations

Don’t forget: Cisco Live sessions will be available for viewing on-demand after the event at www.ciscolive.com/global/on-demand-library/.

Continue Your Education

- Demos in the Cisco campus
- Walk-in Self-Paced Labs
- Tech Circle
- Meet the Engineer 1:1 meetings
- Related sessions

Thank you

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R&S Related Cisco Education Offerings

Course	Description	Cisco Certification
CCIE R&S Advanced Workshops (CIERS-1 & CIERS-2) plus Self Assessments, Workbooks & Labs	Expert level trainings including: instructor led workshops, self assessments, practice labs and CCIE Lab Builder to prepare candidates for the CCIE R&S practical exam.	CCIE® Routing & Switching
• Implementing Cisco IP Routing v2.0 • Implementing Cisco IP Switched Networks V2.0 • Troubleshooting and Maintaining Cisco IP Networks v2.0	Professional level instructor led trainings to prepare candidates for the CCNP R&S exams (ROUTE, SWITCH and TSHOOT). Also available in self study eLearning formats with Cisco Learning Labs.	CCNP® Routing & Switching
Interconnecting Cisco Networking Devices: Part 2 (or combined)	Configure, implement and troubleshoot local and wide-area IPv4 and IPv6 networks. Also available in self study eLearning format with Cisco Learning Lab.	CCNA® Routing & Switching
Interconnecting Cisco Networking Devices: Part 1	Installation, configuration, and basic support of a branch network. Also available in self study eLearning format with Cisco Learning Lab.	CCENT® Routing & Switching

For more details, please visit: http://learningnetwork.cisco.com
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Design Cisco Education Offerings

Course	Description	Cisco Certification
Designing Cisco Network Service Architectures (ARCH) Version 3.0	Provides learner with the ability to perform conceptual, intermediate, and detailed design of a network infrastructure that supports desired capacity, performance, availability required for converged Enterprise network services and applications.	CCDP® (Design Professional) (Available Now)
Designing for Cisco Internetwork Solutions (DESGN) Version 3.0	Instructor led training focused on fundamental design methodologies used to determine requirements for network performance, security, voice, and wireless solutions. Prepares candidates for the CCDA certification exam.	CCDA® (Design Associate) (Available Now)

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